MATERIALS PRICES AND PRODUCTIVITY

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Abstract

There is substantial within-industry variation, even within industries that use and produce homogeneous inputs and outputs, in the prices that plants pay for their material inputs. I explore, using plant-level data from the U.S. Census Bureau, the consequences and sources of this variation in materials prices. For a sample of industries with relatively homogeneous products, the standard deviation of plant-level productivities would be 7% lower if all plants faced the same materials prices. Moreover, plant-level materials prices are both persistent across time and predictive of exit. The contribution of net entry to aggregate productivity growth is smaller for productivity measures that strip out differences in materials prices. After documenting these patterns, I discuss three potential sources of materials price variation: geography, differences in suppliers' marginal costs, and suppliers' price discriminatory behavior. Together, these variables account for 13% of the dispersion of materials prices. Finally, I demonstrate that plants' marginal costs are correlated with the marginal costs of their intermediate input suppliers.

1 Introduction

There is substantial within-industry variation, even within industries that use and produce homogeneous inputs and outputs, in the prices that establishments pay for their material inputs. This paper assesses the implications and sources of this variation in materials

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prices. When input prices differ across plants, some plants will have low marginal costs not only because they efficiently transform inputs into outputs, but also because they happen to purchase their intermediate inputs at particularly low prices. Using plant-level data from the U.S. Census Bureau, I document that intermediate input price dispersion is substantial, persistent across time, and an important factor in explaining dispersion in measured total factor productivity. I then provide suggestive evidence that plants' materials prices[1] are inversely related to their suppliers' productivities, and that intermediate input suppliers engage in price discriminatory behavior. Lastly, I relate plants' productivities to those of their suppliers. I show that these patterns can be matched with a heterogeneous-establishment industry model in which output and factor markets are imperfectly competitive.

Large, persistent, within-industry productivity differences motivate my analysis. Syverson (2004b), for example, estimates that, in the average 4-digit manufacturing industry, the 90th percentile plant has a total factor productivity that is approximately 90% higher than the 10th percentile plant.[2] Given the importance that a plant's productivity has on its growth and survival, as well as the strong relationship between countries' GDPs and the average productivities of their firms,[3] several papers have tried to explain why some plants are productive while others are not. This literature has argued that relatively productive plants are more likely to: employ high quality inputs (Sakellaris and Wilson (2004) and Fox and Smeets (2011)), patent (Balasubramanian and Sivadasan (2011)), expose themselves to export or import markets (Bernard and Jensen (1999), Eslava et al. (2004, 2009), and Halpern, Koren, and Szeidl (2011)), and use best-practice management techniques (Bloom and Van Reenen (2010)). In addition, productivity dispersion is larger in markets with less intense competition (Syverson (2004a)) and in countries with larger factor misallocations (Hsieh and Klenow (2009)).

In the cited studies, plants' productivities are calculated as the ratio of outputs to inputs. Usually, data on input and output prices are not collected, meaning that–in most cases–real revenues are the measure of establishment outputs, while real input expenditures are the measure of establishment inputs.[4] With these measures, an establishment's measured

[1] Throughout this paper, I will use the terms "intermediate inputs" and "materials" interchangeably.

[2] Large within-industry productivity differences have been documented across a wide range of sectors and countries. See Bartlesman and Doms (2000) and Syverson (2011) for surveys.

[3] Prescott (1998), for example, argues that cross-country differences in (physical or human) capital stocks can only explain a small fraction of cross-country per capita GDP differences. The residual explanation for GDP per capita differences is that firms' TFP levels vary across countries.

[4] Five partial exceptions are Syverson (2004a), Eslava et al. (2004, 2009), Hsieh and Klenow (2009), and Ornhagi (2006).

In Syverson (2004a) and Hsieh and Klenow (2009), the authors utilize establishment-level output price data, but do not have establishment-level intermediate input price data. Ornhagi (2006), on the other hand, has data on materials prices. His analysis, focuses on the estimation of input elasticities, instead of the

productivity will depend on the output price it is able to charge and the factor prices it is charged. Potentially, an establishment's measured productivity may have no relationship with how efficient it is in transforming inputs into outputs. Quantifying the extent to which input price variation confounds the measurement of plants' technical efficiencies is one of the main contributions of the this paper.[5]

In empirically documenting a positive relationship between buyers' and suppliers' marginal costs, this paper contributes to two additional literatures. The first set of papers contain models that predict, but have not empirically confirmed, that firms/plants of similar productivities tend to trade with one another. Kremer (1993), for example, develops a model in which the productivity of different production units within a sequential production process is complementary. This model predicts that, as a result of the complementarity, units with similar productivity levels will choose to interact with one another. Oberfield (2011) constructs a model in which buyers meet supplier contacts stochastically, over time; each buyer chooses the supplier, among its contacts, that allows it to produce at the lowest marginal cost. In this model, as well, linked plants' measured productivities will be positively correlated.

Papers in the second literature explore the extent to which shocks to individual firms may have aggregate implications. An unknown, key parameter in these papers is the strength of the relationship between buyer and seller productivity. When buyer and seller productivities are strongly correlated, small differences between countries, in firm-level productivities, will lead to large differences in countries' GDPs (Jones (2011)). Similarly, when buyer and seller productivities are strongly correlated, shocks to a small set of firms will foster large aggregate fluctuations (Acemoğlu et al. (2011; Theorem 3)).

In the following section, Section 2, I sketch out a model in which plants sell output to a representative consumer and purchase intermediate inputs from suppliers in an upstream sector. The model is designed to match the observed correlations between plants' input

distribution of plant-level productivities, which is the focus here.

Perhaps closest to the current paper, Eslava et al. (2004, 2009) use plant-level input and output-price data from Colombia to test the hypothesis that a trade liberalization stiffens the competitive environment, forces low-productivity plants to exit, and thus increases aggregate productivity.

Among these papers, only Syverson (2004a) restricts the sample to homogeneous-output industries. So, some of the variation in quantity total factor productivity in Eslava et al. (2004, 2009) and Hsieh and Klenow (2009) will be a result of differences in output or input quality.

[5]The potential confounding effects of input and output price variation in productivity estimation are already well known. Both Katayama, Lu, and Tybout (2009) and Gorodnichenko (2010) argue, in detail, how plant-level measured productivities may have little to do with plants' technical efficiencies. After arguing that conventional productivity estimates may be misleading, these papers propose structural estimators of establishments' cost and revenue functions, exploiting information derived from the solutions to their cost minimization and/or profit maximization problems. The results contained in the current paper complement and motivate the analysis of Katayama, Lu, and Tybout (2009) and Gorodnichenko (2010).

prices, output prices, and different productivity measures, as well as the correlation between plants' suppliers' marginal costs and plants' own materials prices. In the model, establishments enjoy some degree of monopoly power in output markets and monopsony power in factor markets. As in Foster, Haltiwanger, and Syverson (2008), imperfect competition in the output market yields a positive relationship between revenue productivity and output prices and a negative relationship between quantity productivity and output prices. What is new, compared to most other models of heterogeneous-establishment industries, is a discussion of how intermediate input prices are determined.[6] I assume that each plant has access to only a small set of potential suppliers. Plants that are fortunate to be connected to low-marginal cost suppliers pay lower-than-average materials prices. Because buyers and suppliers of the intermediate input share the gains from the buyer-supplier relationship, the marginal costs of linked counterparties will correlated with one another.

I begin Section 3 by describing the two plant-level datasets—the Census of Manufacturers and the Commodity Flow Survey—employed in this article, as well as the set of industries that comprise my sample. Building heavily off of Foster, Haltiwanger, and Syverson (2008), I focus on the few industries–such as gasoline, ready-mix concrete, and cardboard boxes–for which plants' output prices and materials prices can be computed and meaningfully compared across establishments. In Section 3.3, I specify five assumptions that allow me to compute plant-level productivity measures. As much as possible, I choose assumptions that are common in the literature on estimation of plants' productivities. The one key difference with the literature is a relaxation of the assumption that all plants within each industry purchase their intermediate inputs at the same price.[7] In Section 3.4, I define three different measures of plant productivity. These three measures differ according to their inclusion or exclusion of plant-level prices. Revenue total factor productivity (TFPR) is computed using industry-level price indices for both plants' outputs and intermediate inputs. Quantity total factor productivity (TPFQ) uses plant-level output prices, but industry-level price indices for intermediate inputs. Finally, technical efficiency (which I denote Φ) uses both plant-level materials and output prices.

I establish, in Section 4.1, the hypothesized negative correlation between TFPQ and materials prices. In my benchmark sample, the correlation between the logarithm of TFPQ and materials prices is -37%. In Section 4.2, I compute the fraction of TFPQ dispersion

[6] Several papers model the allocations and profits that emerge out different networks of buyer-supplier relationships. Three examples are de Fontenay and Gans (2007), Manea (2011), and Oberfield (2011). Unlike these papers, the analysis in Section 2.2 invokes convenient assumptions on suppliers' marginal costs and the method by which prices are determined. In combination, these assumptions yield relatively simple expressions for the distribution of transaction prices.

[7] However, I maintain the conventional assumption that there is no within-industry dispersion in the prices at which plants purchase their capital; see Section 3.3.

that is due to differences in materials prices: the standard deviation would be 7% lower, and the 75/25 ratio would be 10% lower, in a counterfactual world in which all plants face the same materials prices. To give context, 7% to 10% is larger than the fraction of productivity dispersion explained by the competitive environment (Syverson (2004a)), but smaller than fraction explained by differences in labor quality (Fox and Smeets (2011)).

As I demonstrate in Sections 4.3 and 4.4, plant-level intermediate input prices are both persistent across time and important predictors of plant survival. The 1-year autocorrelation of the logarithm plants' material prices is 80%. Thus, the serial correlation of the logarithm of input prices is comparable to that of the logarithm of TFPR, TFPQ, or output prices, which are 73%, 71%, and 79%, respectively. In addition, plants that pay more for their material inputs are also significantly more likely to exit the industry: a one standard deviation increase in plants' materials prices corresponds to a 1.1 percentage point increase in the probability that the plant will exit within the next five years. At the same time, there seems to be no difference between entrants and incumbents in the prices that they pay for intermediate inputs. Following from the negative correlations between quantity productivity and input/output prices, the productivity advantage of entrants (compared to incumbents) and surviving plants (compared to exiting plants) is highest when using TPFQ, and lower when using either TFPR or Φ, as the productivity measure. Concomitantly, the contribution of net entry to aggregate productivity growth is smaller for productivity measures that embody plants' output prices (i.e., TFPR, but not TFPQ or Φ), but larger for productivity measures that embody input prices (i.e., TFPR and TFPQ, but not Φ).

In Section 5, I address the question of why materials prices differ within a given industry. An analysis of the sources of materials price dispersion is critical to understanding plants' contributions to social welfare. Plants that, for example, have low materials prices from exploiting monopsony power are contributing nothing to social welfare. On the other hand, plants that search for low-marginal-cost suppliers are contributing to social welfare, even though having low-marginal-cost suppliers do not increase plants' own technical efficiencies.

I offer three potential explanations for within-industry differences in materials prices. First, plants in particular geographic regions enjoy particularly low input prices due, for example, to the abundance of primary materials with which the intermediate input is produced. Second, plants will pay relatively little for their intermediate inputs when their suppliers are exceptionally productive: productive upstream plants are passing on some of their low marginal costs through to their buyers. Thirdly, suppliers tend to charge different prices for their outputs across different destinations. I take this last finding as evidence of price discriminatory behavior, a third source of price variation in intermediate goods markets. For a

pooled sample of ready-mix concrete and cardboard box manufacturers, these three sources reduce the observed price variation by 13%. In terms of explaining variation in materials prices, both the across-supplier component (low-marginal-cost suppliers charge, on average, low prices) and the within-supplier component (a given supplier will charge different prices to different downstream plants) are consequential.

Section 6 concludes. Proofs (Appendix A), other useful formulas (Appendix B), a more detailed description of the construction of the benchmark sample (Appendix C), robustness checks (Appendix D), and bootstrapped confidence intervals (Appendix E) are all relegated to appendices.

2 Theoretical Motivation

In this section, I motivate the empirical analysis of Sections 4 and 5, using an amended version of the model given in Section 2 of Foster, Haltiwanger, and Syverson (2008). The model consists of an industry of plants, which sell output to a representative consumer, and purchase intermediate inputs from an upstream industry.

The analysis consists of two, largely independent, parts. In Section 2.1, the prices that plants pay for intermediate inputs are exogenously specified. Besides facing different materials prices, plants are heterogeneous in their technical efficiencies and appeal to consumers. The main result of this subsection is Proposition 2, which contains a series of relationships among plants' productivity measures, input prices, and output prices.

Section 2.2 describes how materials prices are determined. In the model, plants have access to different sets of heterogeneous-cost suppliers. Plants share, with their suppliers, the surplus generated by the buyer-supplier relationship. As a result, plants with low-marginal-cost suppliers will purchase their intermediate inputs more cheaply, generating a positive relationship between the marginal costs of buyers and suppliers.

2.1 Equilibrium conditions, taking materials prices as given

The analysis focuses on an industry, with $\tilde{I}$ plants, each producing a differentiated good.

The expenditure function of the representative consumer is given by:[8]

$$\log \Upsilon = \tilde{\Delta}_0 + \sum_{i=1}^{\tilde{I}} \left(\tilde{\Delta}_i \log P_i^{out} - \frac{\gamma\left(\tilde{I}-1\right)}{\tilde{I}} \left(\log P_i^{out}\right)^2 \right) + \frac{1}{2}\frac{\gamma}{\tilde{I}} \sum_{i=1}^{\tilde{I}} \sum_{k=1,k\neq i}^{\tilde{I}} \log P_i^{out} \cdot \log P_k^{out} \tag{1}$$

In this equation, Υ is the total revenues of plants in the industry, P_i^{out} is the price that plant i charges for its output, $\tilde{\Delta}_0$ is a demand shifter for the industry's output, and $\tilde{\Delta}_i$ are idiosyncratic demand shocks for the output of plant i.[9] The parameter γ (restricted to be greater than 0) reflects the degree to which the representative consumer is able to substitute across different plants' outputs.

In general, some subset of the $\tilde{I}$ plants may decide to not produce, in equilibrium. (Below, I will provide an inequality describing when plants are able to profitably produce.) Without loss of generality, label plants so that only the first $I \leq \tilde{I}$ plants produce. Then, the representative consumer's expenditure function becomes (see Theorem 1 of Bergin and Feenstra (2009)):

$$\log \Upsilon = \Delta_0 + \sum_{i=1}^{I} \left(\Delta_i \log P_i^{out} - \frac{\gamma\left(I-1\right)}{I} \left(\log P_i^{out}\right)^2 \right) + \frac{1}{2}\frac{\gamma}{I} \sum_{i=1}^{I} \sum_{k=1,k\neq i}^{I} \log P_i^{out} \cdot \log P_k^{out} \tag{2}$$

$$\text{where } \Delta_0 = \tilde{\Delta}_0 \cdot \left(\sum_{i=I+1}^{\tilde{I}} \left(\tilde{\Delta}_i\right)^2 - \frac{1}{I}\left(\sum_{i=I+1}^{\tilde{I}} \tilde{\Delta}_i \right)^2 \right) \quad \text{and } \Delta_i = \tilde{\Delta}_i + \frac{1}{I}\left(1 - \sum_{i=1}^{I} \tilde{\Delta}_i\right)$$

Using Shephard's Lemma, and differentiating Equation 2 with respect to $\log P_i^{out}$, yields a simple expression for R_i, the share of industry revenues that go to plant i:

$$R_i = \frac{d \log \Upsilon}{d \log P_i^{out}} = \Delta_i - \gamma \frac{I-1}{I} \log P_i^{out} + \frac{\gamma}{I} \sum_{k=1,k\neq i}^{I} \log P_k^{out} \tag{3}$$

As γ increases, plants' revenues are increasingly responsive to their prices.

I assume that each plant, i, produces with a constant returns to scale Cobb-Douglas

[8] Consumer preferences are specified differently in Foster, Haltiwanger, and Syverson (2008). In that paper, demand for each variety is linear in prices. Linear demand yields predicted correlations among the *levels* of prices and productivities. Since, in Section 4, I will correlate the *logarithm* of plants' input prices, output prices, and productivity measures, I prefer the translog specification: this set of preferences generates predictions on the relationships among the logarithm of prices and productivity measures.

I eschew the simpler CES specification of consumer preferences, as this specification–in combination with monopolistic competition–would generate the counterfactual prediction that revenue total factor productivities are unrelated to marginal costs.

[9] The $\tilde{\Delta}_i$ are restricted to sum to 1: $\sum_{i=1}^{\tilde{I}} \tilde{\Delta}_i = 1$.

production function, with intermediate inputs (N_i) and labor (L_i) as the two inputs:

$$Q_i = \Phi_i \left(\frac{L_i}{1-\sigma}\right)^{1-\sigma} \left(\frac{N_i}{\sigma}\right)^{\sigma} \tag{4}$$

While factor elasticities are the same throughout the industry, each plant has a unique technical efficiency, Φ_i. I assume that the wage, W, is the same for all plants in the industry, but the price of the intermediate input, P_i^{in}, may differ. The process by which P_i^{in} is determined will be specified in the next subsection. For now, it suffices to assume that P_i^{in} is constant in N_i (e.g., there are no quantity discounts). Each plant's marginal cost of production equals:

$$MC_i = \frac{1}{\Phi_i} W^{1-\sigma} \left(P_i^{in}\right)^{\sigma} \tag{5}$$

Given the preferences defined by Equations 1-3 and the production function defined by Equation 4, plants' pricing behavior takes a simple form.

Proposition 1 *When the revenue share, R_i, of each plant is sufficiently close to 0, the output price, P_i^{out}, is approximated by the following equation:*[10]

$$\begin{aligned} \log P_i^{out} &\approx \frac{1}{2}\left(\log MC_i + \overline{\log MC}\right) + \frac{1}{2\gamma}\left(\Delta_i + \overline{\Delta}\right) \\ &\approx \frac{1}{2}\left(\sigma \log P_i^{in} - \log \Phi_i + \sigma\overline{\log P^{in}} - \overline{\log \Phi}\right) + \frac{1}{2\gamma}\left(\Delta_i + \overline{\Delta}\right) \end{aligned} \tag{6}$$

where, for any variable X, $\overline{X} \equiv \frac{1}{I}\sum_{i=1}^{I} X_i$.

Proof. See Appendix A on page 43. ■

Output price variation has three sources: Plants with low output prices are technically efficient (Φ_i is large), have low material prices (P_i^{in} is small), or have low demand shocks (Δ_i is small). Even though low marginal costs are correlated with low output prices, plants with low marginal costs also have larger than average markups.

Using Equation 6, I can write plants' quantity and revenue total factor productivities in terms of Φ, P^{in}, and Δ.

$$\log TFPQ_i \equiv \log\left(\frac{Q_i}{Q_i \cdot MC_i}\right) = \log \Phi_i - \sigma \log\left(P_i^{in}\right) - (1-\sigma)\log W \tag{7}$$

$$\begin{aligned} \log TFPR_i &= \log TFPQ_i + \log P_i^{out} \\ &\approx \frac{1}{2}\left(\log \Phi_i - \overline{\log \Phi} - \sigma \log\left(P_i^{in}\right) + \sigma\overline{\log\left(P^{in}\right)} + \frac{1}{\gamma}\left(\Delta_i + \overline{\Delta}\right)\right) \end{aligned} \tag{8}$$

[10] Equation 6 is derived by invoking the approximation $\log(1+R_i) \approx R_i$, which becomes more accurate as $R_i \to 0$.

Φ, which embodies plants' ability to transform inputs into outputs, is independent of conditions in factor and output markets. $TFPQ$ reflects both the ability to transform inputs into outputs, as well as conditions in factor markets. Finally, a plant's $TFPR$ is determined by its technical efficiency, conditions in factor markets, and conditions in output markets.

Plugging Equation 6 into Equation 3 yields the following expression for the revenue share of plant i:

$$R_i \approx \frac{1}{2}\left(\Delta_i + \overline{\Delta}\right) - \frac{\gamma}{2}\left(\log MC_i - \overline{\log MC}\right) \tag{9}$$

In combination, Equations 8 and 9 imply that a plant will have a positive revenue share ($R_i > 0$) if and only if its $\log TFPR$ is greater than 0.

Finally, for plants $i \in \{1, ..., I\}$, profits equal:

$$\begin{aligned} \Pi_i &= \frac{\Upsilon \cdot R_i}{P_i^{out}}\left(P_i^{out} - MC_i\right) \\ &\approx \Upsilon\left[\frac{1}{2}\left(\Delta_i + \overline{\Delta}\right) - \frac{\gamma}{2}\left(\log MC_i - \overline{\log MC}\right)\right] \cdot \left[e^{\frac{1}{2}\left(\overline{\log MC} + \frac{\overline{\Delta}}{\gamma}\right)} - \left(MC_i\right)^{1/2} e^{-\frac{1}{2\gamma}\Delta_i}\right] \end{aligned} \tag{10}$$

Plants $i \in \left\{I+1, ..., \tilde{I}\right\}$ earn zero profits. Total industry sales, Υ, can be computed by plugging in the optimal pricing decisions (Equation 6) into Equation 2.[11]

To close the model, and solve for $\tilde{I}$, I assume that there is a large pool of potential entrants, who may enter only after paying a sunk cost. Potential entrants will weigh expected profits to the cost of entry. To compute the expected profitability, one would integrate Equation 10 over the joint distribution, $\varphi(\Delta, P^{in}, \Phi)$, of Δ_i, P_i^{in}, and Φ_i.

Using Equations 6-8, one may express relationships between plants' productivities and input/output prices. These relationships, which are collected in Proposition 2, depend on the relationships among the three exogenous variables (Δ, Φ, and P^{in}). In turn, the correlations between Δ, Φ, and P^{in} will depend on φ as well any correlation that is induced by selection on $TFPR$. Plants will profitably produce if Δ or Φ is sufficiently large, or if P^{in} is sufficiently low. Even if φ is multiplicatively separable in Δ, Φ, and P^{in}, the correlation between Δ and Φ will be negative in the observed sample. Likewise, selection on $TFPR$ will tend to induce a positive correlation between Δ and P^{in}, and a positive correlation between P^{in} and Φ.

[11] One can show that (see Appendix B.1 on page 45):

$$\log \Upsilon = \Delta_0 + \frac{I\gamma}{8}\left[2 \cdot Cov\left(\frac{\Delta}{\gamma}, \log MC\right) + 3 \cdot Var\left(\frac{\Delta}{\gamma}\right) - Var(\log MC)\right] + \frac{1}{2}\left[E\left[\log MC\right] + E\left[\frac{\Delta}{\gamma}\right]\right]$$

.

In the following proposition, I assume that these correlations are not too far from 0 in the observed sample. Without making this assumption, it would be difficult to say anything meaningful about the correlations between plants' productivity measures and input/output prices.

Proposition 2 *Assume that a) Cov($\log P^{in}$, $\log \Phi$), Cov($\log P^{in}$, Δ), and Cov(Δ, $\log \Phi$) are sufficiently close to 0 in the observed sample, and b) $R_i \to 0$ for all plants i. Then, the following relationships hold:*

1. $Cov(\log TFPR, \log TFPQ) = \frac{1}{2}Var(\log TFPQ) + \frac{1}{2\gamma}\left[Cov(\Delta, \log \Phi) - \sigma Cov(\Delta, \log P^{in})\right] > 0$
2. $Cov(\log TFPQ, \log \Phi) = Var(\log \Phi) - \sigma Cov(\log P^{in}, \log \Phi) > 0$
3. $Cov(\log TFPQ, \log P^{in}) = -\sigma Var(\log P^{in}) + Cov(\log P^{in}, \log \Phi) < 0$
4. $Cov(\log TFPR, \log P^{out}) = \frac{1}{4}Var(\log TFPQ) + \frac{1}{4\gamma^2}Var(\Delta) > 0$
5. $Cov(\log TFPQ, \log P^{out}) = \frac{1}{2\gamma}\left[Cov(\Delta, \log \Phi) - \sigma Cov(\Delta, \log P^{in})\right] - \frac{1}{2}Var(\log TFPQ) < 0$
6. $Cov(\log P^{in}, \log P^{out}) = \frac{1}{2}\left[\sigma Var(\log P^{in}) - Cov(\log P^{in}, \log \Phi) + \frac{Cov(\log P^{in}, \Delta)}{\gamma}\right] > 0$
7. $Var(\log TFPQ) - Var(\log \Phi) = \sigma^2 Var\left(\log P^{in}\right) - 2\sigma Cov\left(\log \Phi, \log P^{in}\right) > 0$

In Section 4, I check that these seven predicted relationships hold for a sample of homogeneous-goods industries.[12] Moreover, consistent with Equations 8-9, I document that plants with low technical efficiencies and high input prices tend to exit with greater probability.[13] What is unclear, so far, is why intermediate input prices differ across plants. I develop a theory of materials price dispersion in the following subsection.

[12] As I will show in Section 4, $TFPQ$ is significantly more disperse than $TFPR$. The model that I have presented in this subsection yields an ambiguous prediction on the relative dispersions of $TFPQ$ and $TFPR$:

$$\begin{aligned} Var(\log TFPQ) - Var(\log TFPR) &= -Var(\log P^{out}) - 2Cov\left(\log P^{out}, \log TFPQ\right) \\ &= \frac{3}{4}Var\left(\log TFPQ\right) + \frac{3}{4\gamma}Cov\left(\log TFPQ, \Delta\right) - \frac{1}{4\gamma^2}Var\left(\Delta\right) \\ &\lesseqgtr 0 \end{aligned}$$

[13] To develop this prediction rigourously, I would need to construct a model–similar to that of Asplund and Nocke (2006)–with multiple periods, in which Φ, Δ, and P^{in} evolve stochastically.

2.2 Materials prices

There are two aims of this subsection. The first is to show that plants' marginal costs are positively related to those of their suppliers: plants that are lucky enough to be connected to an exceptionally low-cost supplier will purchase their intermediate inputs at a low price, and thus have low marginal costs. The second aim of this subsection is to decompose materials price variation into a between-supplier and a within-supplier component. Between-supplier variation in materials prices reflects differences in marginal costs, across suppliers, whereas within-supplier variation occurs because of differences in the bargaining position–determined by the web of buyer-supplier relationships–between buyers and suppliers.

I assume that manufacturers of the intermediate input, the upstream plants $j \in \{1, ..., J\}$, produce at constant marginal cost, MC_j. The intermediate input produced by the upstream industry plants are not vertically differentiated: the downstream plant will choose the supplier that provides the input at the lowest price. Each downstream plant, i, has access to $\xi \geq 2$ potential suppliers. While the number of potential suppliers is the same, the identities of these potential suppliers differ across the downstream plants.[14] Let $MC_{i(k)}$ denote the marginal cost of the kth most efficient supplier of plant i. Finally, let $G(\cdot)$ denote the cumulative distribution (with associated probability distribution function, $g(\cdot)$) corresponding to the marginal cost of a randomly drawn supplier.

Suppliers engage in Bertrand competition for the patronage of each downstream plant, i. Given this assumption, each buyer purchases from the lowest-marginal-cost supplier at price $P_i^{in} = MC_{i(2)}$. The p.d.f. of the 0materials price, P^{in}, conditional on the marginal cost of the supplier, MC^{up}, equals:[15]

$$h(P^{in}|MC^{up}) = (\xi - 1)\frac{g\left(P^{in}\right)}{1 - G(P^{in})}\left(\frac{1 - G(P^{in})}{1 - G(MC^{up})}\right)^{\xi - 1} \text{ for } P^{in} > MC^{up} \tag{11}$$

By integrating Equation 11, moments of any function, $f(\cdot)$, of the materials price that i pays, conditional on the marginal cost of the best potential supplier, are given by the

[14] The idea that plants are exogenously restricted in whom they can potentially trade with is taken from de Fontenay and Gans (2007), Manea (2011), and Oberfield (2011). The restriction may have any number of origins. For example, the set of counterparties that each plant can trade with may be the result of a costly search process, as outlined in Chaney (2011).

[15] In Appendix A, I show that $H(P^{in}|MC^{up}) = 1 - \left(\frac{1-G(P^{in})}{1-G(MC^{up})}\right)^{\xi-1}$. Differentiating this expression, with respect to P^{in}, gives the desired result.

following expression:

$$E\left[f\left(P^{in}\right)|MC^{up}\right] = \frac{\xi - 1}{\left(1 - G(MC^{up})\right)^{\xi-1}} \int_{MC^{up}}^{\infty} f(z)g(z)\left(1 - G(z)\right)^{\xi-2} dz \tag{12}$$

According to Equations 11 and 12, the conditional distribution of the materials price shifts to the right as the supplier's marginal cost increases. Combining this finding with the positive relationship between i's marginal cost and its own materials price (Proposition 2, part 3) yields a positive relationship between the marginal costs of establishments that trade with one another. These two findings are stated in the following proposition:

Proposition 3 *Under the same assumptions as in Proposition 2:*

1. *The materials price is increasing, in the sense of first order stochastic dominance, in the marginal cost of the supplier.*

2. *Plants' marginal costs are positively correlated to their suppliers' marginal costs.*

Proof. See Appendix A on page 44. ■

Proposition 3 generates qualitative predictions on the relationships between plants' marginal costs, their materials prices, and their suppliers' marginal costs. Next, I will argue that the predicted price dispersion is substantial, and that both the within-supplier and across-supplier components (which I will define below) are important drivers of the observed variation in materials prices. One can show, for future reference, that the probability distribution functions of the first and second order statistics are:

$$h_1(x) = \xi \cdot g(x) \cdot \left(1 - G(x)\right)^{\xi-1} \tag{13}$$

$$h_2(y) = (\xi - 1) \cdot G(y) \cdot g(y) \cdot \left(1 - G(y)\right)^{\xi-2} \tag{14}$$

I assume that suppliers' quantity productivities are drawn from the Pareto distribution, with shape parameter, ζ.[16] Then, for $MC \in [0, 1]$, $G(MC) = MC^{\zeta}$ and $g(MC) = \zeta MC^{\zeta-1}$.[17]

In the left panel of Figure 1, I plot $SD\left(\log P^{in}\right)$, which is computed by integrating over the probability distribution function of prices, as given in Equation 14. For all values of ξ, the variance of observed prices is of the same order of magnitude as the variance of the potential suppliers' marginal costs.

[16] The proceeding results are similar under the assumption of lognormal quantity productivities.

[17] Taking 1 to be the upper limit of the support of potential suppliers' marginal costs is only a normalization. One could re-define the units of the intermediate good so that the highest-possible marginal cost of the supplier is any positive constant.

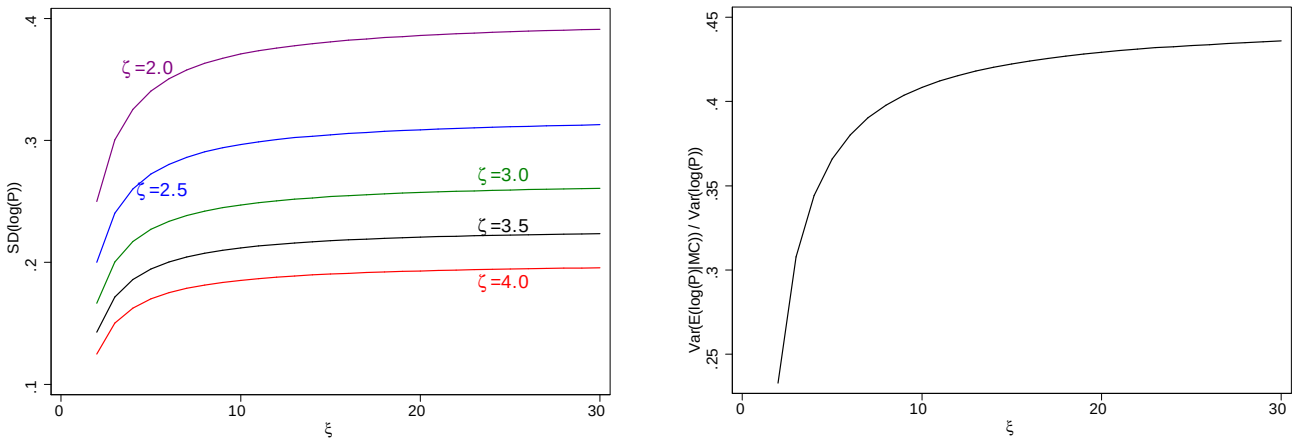


Figure 1: Theoretical results. The left panel gives the standard deviation of $log(P^{in})$, for different combiantions of ζ and ξ. The right panel displays the fraction of materials price variation that originates from the across-supplier component.

The right panel of Figure 1 displays the fraction of materials price variation that originates from the differences in suppliers' marginal costs.[18] Variation in materials prices, $Var\left[\log\left(P^{in}\right)\right]$, equals the sum of $Var\left[E\left[\log\left(P^{in}\right)\middle|MC^{up}\right]\right]$ and $E\left[Var\left[\log\left(P^{in}\right)\middle|MC^{up}\right]\right]$. $Var\left[E\left[\log\left(P^{in}\right)\middle|MC^{up}\right]\right]$ reflects the variation in materials prices due to the differences in MC^{up} that occur across upstream plants, while $E\left[Var\left[\log\left(P^{in}\right)\middle|MC^{up}\right]\right]$ reflects the variation in prices due to differences in the relative bargaining position of the buyers and sellers of the intermediate good. The takeaway from this figure is that, for all values of ξ, both the within and between-supplier components contribute substantially to the observed variation in materials prices.

In summation, when buyers and suppliers have limited trading opportunities, plants with access to low-marginal-cost suppliers will pay below average prices for their intermediate inputs. This, in turn, generates a positive correlation between the measured productivities of plants that transact with one another. In a similar vein, materials prices will be spatially correlated to the extent that upstream plants' marginal costs can be explained by geography. While a large fraction of materials price dispersion can be explained by the marginal cost of the supplier, suppliers also price discriminate across their customers, based on the downstream plants' outside options. These findings are verified in Section 5: I show that plants' materials prices are, indeed, related to the marginal costs of their suppliers, and that individual sellers do sell the same output at different prices to different downstream plants. I also find some evidence that suppliers and buyers' marginal costs are correlated.

[18] I plot only one line, because the fraction of materials price variation from the across-supplier component is unchanged in ζ.

3 Data and Definitions

3.1 Data sources

For this paper, the main data sources are the Commodity Flow Survey and the Census of Manufacturers. Both datasets are collected and maintained by the U.S. Census Bureau.

The Census of Manufacturers is conducted every five years, in years ending in '2' or '7'. This dataset contains information on manufacturing establishments' productive characteristics, including its employment of production and nonproduction workers, measured in hours; the book value of its building and machine capital; and expenditures on electricity. For certain industries, establishments with five or more employees are asked additional questions about the products that they produce and the materials that they consume. Of particular importance for the current paper, establishments list both the quantity and the value of each of the products they produce (at the 7-digit level) and the quantity and value of each of the materials they consume (at the 6-digit level).[19] For this paper, I use the 1972, 1977, 1982, 1987, 1992, and 1997 versions of the Census of Manufacturers.

The Commodity Flow Survey is necessary to impute buyer-supplier relationships, as I do in Section 5.1. The Commodity Flow Survey was first conducted in 1993, and thereafter in years ending in '2' or '7'. Surveyed establishments are asked to list 25-40 shipments that they make each quarter.[20] Each observation includes information on: the weight and value of the shipment; a five-digit code, specifying the commodity that was shipped; the method of transport (air, truck, rail, courier service, etc...); and the destination zip code. Finally, the Census uses the same plant-level identifier in the Commodity Flow Survey and the Census of Manufacturers. This allows me to combine the information on production and input utilization from the Census of Manufacturers with information on shipment patterns from the Commodity Flow Survey. In Section 5, I employ the 1993 and 1997 Commodity Flow Surveys.

[19] To give the reader some idea of the scope of a 7-digit product, ready-mix concrete (3273000) is one of the larger product groups. Some of the smaller product-groups include corrugated boxes, used as containers of chemicals and drugs, including paints, varnishes, cosmetics, and soaps (2653021); or self-rising family white flour (2044126). For 1992, a description of the product codes can be found at http://www.census.gov/prod/2/manmin/mc92-r-1.pdf.

[20] Not all manufacturers are surveyed in the Commodity Flow Survey. In 1993, approximately 60 thousand (out of the 350 thousand existing manufacturing plants) were surveyed, while, in 1997, approximately 30 thousand plants were surveyed.

3.2 Sample

In this subsection, I describe the construction of the sample. Further details are provided in Appendix C on page 46.

Similar to Roberts and Supina (1996) and Foster, Haltiwanger, and Syverson (2008), the analysis centers around industries for which outputs and inputs are relatively homogeneous. In industries with heterogeneous inputs or outputs, differences in quality may be a main source of the variation in the prices that different firms charge. I would like, as much as possible, to rule out quality as a source of input or output price variation. An additional restriction is that both the inputs and outputs should be measured in units that are comparable across establishments.[21]

The ten industries[22] that comprise the main sample are packaged milk, white wheat flour, ready-mix concrete, grey cotton yarn, cardboard boxes (with the years 1972-1987 and 1992-1997 analyzed separately), gasoline, ground coffee, raw cane sugar, and bulk milk; see Table 1.[23,24] Approximately one-third of the plant-year observations are from plants that manufacture ready-mix concrete. However, when observations are weighted by their real sales, the gasoline industry is the most prominent: approximately three-quarters of the total revenues are earned by plants from this industry.

In addition to producing a product from one of these ten industries, to be in the benchmark sample, the manufacturers must fill out the materials and production supplements. These supplemental forms, which the Census sends out to larger establishments, are necessary to compute the unit values of manufacturers' outputs and materials purchases.

Thus, there are two sources of sample selection. First, I have chosen industries based on the characteristics of the outputs they produce and the inputs they purchase. These industries tend to use intermediate materials more intensively than the broader manufacturing sector. Since the scope for material price differences to cause measured productivity disper-

[21] This second restriction rules out industries like oak, hardwood rough lumber (7-digit product code= 2421163). For this industry, output is measured in units of board feet, but different plants manufacture lumber with different thickness. For this reason, it is difficult to compare different plants' output prices, productivities, or other plant-level characteristics.

[22] When referring to the different subsamples of the benchmark sample, I will use the terms "industry" and "product" interchangeably.

[23] A problem similar to the one described in footnote 21 exists for the post-1992 cardboard box industry. Beginning in 1992, the units of output switch from thousands of pounds to thousands of square feet. I detail my response to this potential problem in Appendix C.

[24] Cardboard boxes, raw cane sugar, gasoline, ground coffee, and ready-mix concrete are included in both the current paper and Foster, Haltiwanger, and Syverson (2008). I could not include carbon black, block ice, or processed ice, as there were insufficiently many plants that filled out both the production and materials supplements. I do not include hardwood flooring or plywood, the last two industries that Foster, Haltiwanger, and Syverson (2008) include; large output price dispersions seem to indicate that the outputs of these industries are not sufficiently homogeneous.

Sample	Units of Output	Material Inputs	N
Milk-Bulk	1000 lbds.	Unprocessed whole milk (88%)	127
White Wheat Flour	50 lbd. sacks	Wheat (90%)	503
Ready-Mix Concrete	1000 cubic yards	Cement (53%), Sand/Gravel (28%)	3708
Carded Cotton Yarn	1000 lbds.	Cotton Fibers (80%), Polyester tow (10%)	431
Boxes, Year$\leq$1987	Short tons	Paper/paperboard (90%)	1820
Gasoline	1000 barrels	Crude petroleum (84%)	692
Ground Coffee	1000 lbds.	Green coffee beans (80%)	300
Raw Cane Sugar	Short tons	Sugar cane (93%)	177
Boxes, Year$\geq$1992	Square feet	Paper/paperboard (89%)	646
Milk-Packaged	1000 quarts	Unprocessed whole milk (72%)	2099
Pooled	-		10503

Table 1: Description of the ten industries in the benchmark sample. The percentages that appear in the Material Inputs column are the fraction of intermediate input expenditures that go to each particular material input. The Material Inputs column shows the inputs that represent greater than 6% of the average plant's total material purchases.

sion increases with the intensity of intermediate input usage (see Equation 22), it is likely that the decline in total factor productivity dispersion is larger for the ten industries in my sample than for the broader manufacturing sector.

Second, the plants that are the focus of this study tend to be larger and more productive, compared to the other plants from their respective industries. The average plant in my sample employs roughly 5 times more employees, and has revenues that are 4 times larger, compared to the average plant of their respective industry. (For more details, see Table 16, in Appendix C). Furthermore, since the probability of exit tends to decrease with size, the plants in my benchmark sample are relatively more likely to survive: plants in the benchmark sample have a 5-year survival rate of 86%, compared to the average survival rate for plants in their industries, 72%.

These sample selection issues limit the generalizability of the results given in Sections 4 and 5. However, by sacrificing generality, I am able to isolate the effect of differences in materials prices on intra-industry productivity dispersion.

3.3 Assumptions

I make five assumptions regarding plants' production technologies and the way in which labor, capital, and electricity are supplied. The dual objectives of these assumptions are to a) take the predictions of Section 2's model to the data, and to b) highlight the importance of material price dispersion in the measurement of plant-level productivities. With these goals in mind, I will, as much as possible, adhere to conventional assumptions made in the literature on plant-level production function estimation.

Assumption 1: Plants within an industry have Cobb-Douglas production functions, with labor, capital, electricity, and materials as the inputs. Furthermore, the factor shares are common across all plants within an industry-year combination. Also, plants' production functions exhibit constant returns to scale.

There are three components to the first assumption: a unitary elasticity of substitution, common factor shares within an industry, and constant returns to scale. The unitary elasticity of substitution is common in studies of plants' production functions, mainly for convenience. However, several authors have estimated an elasticity of substitution between labor and capital that is less than 1 (see Raval (2011)). For the objects of interest, the Cobb-Douglas assumption seems to have little effect on the dispersion of measured productivity. I show, in Appendix D.4 on page 56, that the results of Section 4 are robust to complementarities between material inputs and other inputs. Furthermore, when measuring the fraction of productivity dispersion that can be explained by differences in labor quality, Fox and Smeets (2011; Section 5) find that productivity dispersions are remarkably similar, whether the production function is assumed to be Cobb-Douglas or translog.

The other parts of Assumption 1 are also rather innocuous. In Syverson (2004b), the relationships between within-industry productivity dispersion and other industry characteristics is robust to using plant-specific factor shares when estimating plants' TFPs.[25] Related to the constant-returns-to-scale component of Assumption 1, Syverson (2004a) estimates that the returns to scale are indistinguishable from 1 for plants in the ready-mix concrete industry, the industry that contains roughly one-third of the plants in my sample.[26]

Assumption 2: The unit input costs of capital and electricity are the same for all plants within an industry-year combination. In addition, the marginal cost of each of these inputs is constant in the amount purchased.

[25] Syverson does find, however, that the measured within-industry differences in TFP are smaller when computed using plant-specific factor shares.

[26] Baily, Hulten, and Campbell (1992) estimate returns to scale for a broader set of industries and find the same result.

Data limitations necessitate the assumption that all plants face the same costs for a unit of capital services. The assumption that electricity costs are the same across plants within an industry can be relaxed, without changing any of the results of Section 4.[27] Since I can directly observe the quantity of labor input, I do not need to assume that wages are the same across plants.[28]

Assumptions 1 and 2 are consistent with the set-up of Section 2's model. Assumptions 3-5 deal with the fact that plants may produce multiple outputs and purchase multiple intermediate inputs. These assumptions were not addressed in Section 2, since plants were assumed to produce and purchase only one type of good.

Assumption 3: The fraction of each input employed in producing a particular product equals the plant's share of revenue coming from that product.

The need for Assumption 3, an assumption also made by Foster, Haltiwanger, and Syverson (2008), stems from a limitation of the dataset. In particular, for plants that produce multiple goods, it is impossible to know exactly how much of each input that is used in the production of each output. I make the simplest possible assumption, and assume that each input is allocated in proportion to plant's sales of each product. For example, for a hypothetical plant that employs L units of labor and sells Y_g dollars of good g, for $g \in \{1, ...G\}$, the amount of labor used in the production of good g is

$$L \cdot \frac{Y_g}{\sum_{\hat{g}=1}^{G} Y_{\hat{g}}}. \tag{15}$$

Similar to Foster, Haltiwanger, and Syverson (2008), I argue that the dispersion of productivity is robust to the way in which inputs are allocated to outputs, mainly because the plants in my sample tend to be heavily specialized in the goods they manufacture.

In addition to Assumptions 1-3, which are common in papers that estimate plant-level productivities, I make two assumptions on the substitutability among different material

[27] Davis, Grim, and Haltiwanger (2008) compute plant-level energy prices and show that there is substantial variation, within industries, in the cost of a kilowatt-hour of electricity. In an unreported robustness exercise, I check that the results of Section 4 are virtually identical after relaxing the assumption that all plants face the same electricity prices, the reason being that the expenditure share of energy is small (on average, 2.5%) for plants in the benchmark sample.

[28] Differences in labor quality, across plants, may muddle the interpretation of plants' productivities. Using hours worked as the measure of labor means that plants with exceptionally skilled workers would appear to be highly productive. If workers' wages reflect differences in worker skill (as opposed to, for example, workers' bargaining power), it would be preferable to measure labor inputs by the wages paid by each plant. In an unreported robustness check, I reproduce Tables 2 and 3 using the wage bill, instead of hours worked, as the measure of labor inputs. The results are essentially identical when using this different measure of labor inputs.

inputs. Together, Assumptions 4 and 5 will allow me to compute plant-specific material prices from the data at hand. While restrictive, they are much less so than the common presumption that all plants face the same intermediate input prices.

Assumption 4: If multiple intermediate inputs are observed, the elasticity of substitution between the materials is 0.

This assumption is pertinent only for the two industries, plants producing ready-mix concrete or yarn, for which I observe multiple intermediate inputs being employed. I show, in Appendix D.3 on page 55, that the level of productivity dispersion is extremely robust to moderate levels of substitutability among the different intermediate inputs.

Assumption 5: The elasticity of substitution, between plants' "priced" and "non-priced" materials is 1. In addition, the elasticity of substitution between "non-priced" materials and capital, labor, and materials is also 1.

Here, "priced materials" are the materials that most plants in the industry purchase. For instance, in the case of yarn manufacturers, cotton fibers and polyester tow are the "priced materials." The non-priced materials are purchased by only a few plants in the industry. Again, turning to the yarn industry, approximately 10% of the expenditures on intermediate inputs go to purchases of materials other than cotton fibers (see the 'Material Inputs' column of Table 1). Some of these yarn-producing plants purchase silk fibers; others purchase nylon tow. Since only a few plants purchase these materials, it is difficult to ascertain if plants are purchasing these inputs relatively cheaply or expensively. I treat the "non-priced" materials as if they were any other input for which I do not observe unit prices, such as capital or electricity, and assume that there is a unitary elasticity of substitution between "non-priced" materials and "priced" materials, labor, capital, and electricity.

3.4 Definitions

In this subsection, I define plants' materials and output prices, as well as the three plant-level productivity measures: $TFPQ$, $TFPR$, and Φ. The first two productivity measures are exactly as in Foster, Haltiwanger, and Syverson (2008). The productivity measure that is new to this paper, Φ, aims to isolate plants' abilities to transform inputs into outputs. In particular, this measure, Φ, should not reflect plants' abilities to sell their output at a relatively high price, or to purchase their intermediate inputs at a relatively low price.

I begin by defining plants' input and output prices. The price, P_{ijt}^{out}, that plant i charges for product j in year t is simply the ratio of sales, Y_{ijt}, to physical quantity shipped,

Q_{ijt}:

$$P_{ijt}^{out} \equiv \frac{Y_{ijt}}{Q_{ijt}}. \tag{16}$$

Before defining plant-level input prices, I introduce some notation. Let M_{ijt} be the expenditures on materials of plant i in the production of product j in year t. Plant i's purchases consist of "non-priced" materials, which I denote using M_{ijt}^0, and "priced" materials, which I denote using M_{ijt}^1 (and M_{ijt}^2 if j is produced using two material inputs). Let, $s_{\mu jt}$ denote the average fraction–across plants in my sample in industry j and year t–of materials expenditures that are spent on material μ.[29] Finally, let S_{jt} denote the average fraction of materials expenditures, in industry j in year t, that go to "priced" materials.[30]

For plants in industries that use only one type of "priced" material (i.e., all industries except for ready-mix concrete and yarn manufacturing), the input price is given by the ratio of materials expenditures of the first good, M_{ijt}^1, to the physical quantity, N_{ijt}^1, consumed of the first good.

$$P_{ijt}^{in} \equiv \frac{M_{ijt}^1}{N_{ijt}^1} \tag{17}$$

To construct plant-level materials prices for ready-mix concrete and yarn manufacturers, I begin by defining a unit of the intermediate input bundle as follows:

$$\begin{aligned} N_{ijt} &= \min\left\{\frac{s_{1jt}}{S_{jt}}\frac{N_{ijt}^1}{\bar{N}_{ijt}^1}, \frac{s_{2jt}}{S_{jt}}\frac{N_{ijt}^2}{\bar{N}_{ijt}^2}\right\} \\ &= \lim_{\varrho\to 0}\left(\frac{s_{1jt}}{S_{jt}}\left(\frac{N_{ijt}^1}{\bar{N}_{ijt}^1}\right)^{\frac{\varrho-1}{\varrho}} + \frac{s_{2jt}}{S_{jt}}\left(\frac{N_{ijt}^2}{\bar{N}_{ijt}^2}\right)^{\frac{\varrho-1}{\varrho}}\right)^{\frac{\varrho}{\varrho-1}} \end{aligned} \tag{18}$$

In Equation 18, N_{ijt} is the number of units of the intermediate input bundle purchased by plant i in industry j in year t. Because the units of N_{ijt} have no natural interpretation, it is necessary to normalize by the average input utilization of each of the intermediate goods, $\bar{N}_{ij}^1$ and $\bar{N}_{ij}^2$.[31] Assumption 4 pins down how the two different materials are combined to form the composite intermediate input; relaxing Assumption 4 would involve allowing $\varrho > 0$.

Let $\bar{P}_{ij}^1$ and $\bar{P}_{ij}^2$ be the average unit price of materials 1 and 2 in industry j. Then, the materials bundle's ideal price index equals the value-weighted average of the individual

[29] For example, for j=concrete and μ =cement, $s_{\mu jt}$ would be approximately 0.53, slightly more in some years, slightly less in others.

[30] Continuing with the example from the previous footnote, S_{jt} would be approximately 0.81(= 0.28+0.53) for ready-mix concrete manufacturers.

[31] Klump, McAdam, and William (2011) comprises a discussion on the necessity of normalizing CES production functions when $\varrho \neq 1$. (When $\varrho = 1$, the units can be factored out into a multiplicative constant.)

inputs' prices:

$$P_{ij}^{in} \equiv \frac{s_{1jt}}{S_{jt}} \frac{P_{1ijt}^{in}}{\bar{P}_{1jt}^{in}} + \frac{s_{2jt}}{S_{jt}} \frac{P_{2ijt}^{in}}{\bar{P}_{2jt}^{in}} \tag{19}$$

Having defined plant-level materials and output prices, I am now able to state how plant-level productivities are computed. For each plant, i, producing in industry j, define its quantity total factor productivity ($TFPQ$) in year t as the ratio between the physical quantity produced and the inputs it utilized in the production of this product.[32]

$$TFPQ_{ijt} \equiv Q_{ijt} \left(L_{ijt}\right)^{-\lambda_{jt}} \left(K_{ijt}\right)^{-\kappa_{jt}} \left(E_{ijt}\right)^{-\epsilon_{jt}} \left(M_{ijt}\right)^{-\sigma_{jt}} \tag{20}$$

In Equation 20, L_{ijt}, K_{ijt}, and E_{ijt} denote the amount of labor, capital, and energy used in the production of product j. From Assumption 3, each input listed in Equation 20 equals the total amount purchased of that input multiplied by the share of plant i's revenues derived from producing its main output. As in Foster, Haltiwanger, and Syverson (2008), labor is stated in terms of hours, and capital is computed by summing plants' reported book values of equipment and structures. The industry-year specific cost shares are computed as in Foster, Haltiwanger, and Syverson (2008). Note that, because of Assumption 1, the factor elasticities, λ_{jt}, κ_{jt}, ϵ_{jt} and σ_{jt}, are the same for all plants within an industry-year pair. In addition, $\lambda_{jt} + \kappa_{jt} + \epsilon_{jt} + \sigma_{jt} = 1$ for all j, t pairs. To emphasize, since $M_{ijt} = P_{ijt}^{in} \cdot N_{ijt}$, low materials prices are a factor associated with high $TFPQ_{ijt}$.

Revenue total factor productivity (TFPR) captures a plant's ability to transform a given bundle of inputs into revenue. As Equation 21 makes clear, plants will have a high TFPR for one of two reasons: Either they have high $TFPQ$, or they sell their output at a particularly high price.

$$\begin{aligned} TFPR_{ijt} &\equiv Y_{ijt} \left(L_{ijt}\right)^{-\lambda_{jt}} \left(K_{ijt}\right)^{-\kappa_{jt}} \left(E_{ijt}\right)^{-\epsilon_{jt}} \left(M_{ijt}\right)^{-\sigma_{jt}} \\ &= TFPQ_{ijt} \cdot P_{ijt}^{out} \end{aligned} \tag{21}$$

Finally, plants' technical efficiencies (Φ_{ijt}) purge out the effect of variation in the

[32] There are many ways to estimate plant-level productivity. I would have liked to compare the estimates generated by Equations 20-22 to those computed using other estimation methodologies. Like Foster, Haltiwanger, and Syverson (2008), I am unable to compute plant-level productivities using the methods outlined in Olley and Pakes (1995), Blundell and Bond (2000), Levinsohn and Petrin (2003), and Ackerberg, Caves, and Frazer (2006). These methods generally require annual observations. Unfortunately information on quantities of output produced or intermediate inputs purchased exist only for years in which the Census of Manufacturers is conducted. Most likely, my results would not change if other productivity measures were used. Van Biesebroeck (2008) reports that, unlike estimates of input elasticities, which are sensitive to the estimation methodology, plant-level productivity estimates are highly correlated across different estimation methodologies.

prices that plants pay for their intermediate inputs:

$$\begin{aligned}\Phi_{ijt} &\equiv Q_{ijt} \cdot (L_{ijt})^{-\lambda_{jt}} \cdot (K_{ijt})^{-\kappa_{jt}} \cdot (E_{ijt})^{-\epsilon_{jt}} \cdot \left(M^0_{ijt}\right)^{-\sigma_{jt}\cdot(1-S_{jt})} \cdot (N_{ijt})^{\sigma_{jt}\cdot S_{jt}} \\ &= Q_{ijt} \cdot (L_{ijt})^{-\lambda_{jt}} \cdot (K_{ijt})^{-\kappa_{jt}} \cdot (E_{ijt})^{-\epsilon_{jt}} \cdot (M_{ijt})^{-\sigma_{jt}} \cdot \left(P^{in}_{ijt}\right)^{\sigma_{jt}\cdot S_{jt}} \\ &= TFPQ_{ijt} \cdot \left(P^{in}_{ijt}\right)^{\sigma_{jt}\cdot S_{jt}}\end{aligned} \tag{22}$$

The equality of the first and second lines of Equation 22 follows from Assumption 5: because of the unitary elasticity of substitution between "priced" and "non-priced" materials, $M_{ijt} = \left(M^0_{ijt}\right)^{(1-S_{jt})} \cdot \left(M^1_{ijt} + M^2_{ijt}\right)^{S_{jt}}$. The equality of the second and third lines follows from the definition of $TFPQ$. Equation 22 states that plants will have high $TFPQ_{ijt}$ for one of two reasons: either the plant is technically efficient (Φ_{ijt} is large), or intermediate input prices are low (P^{in}_{ijt} is low).[33]

So that I can compare observations across industries and years, all quantities will be stated relative to the mean for that industry×year. I use lower-case letters to denote the logged, de-meaned values. For any plant-level statistic, X_{ijt}, define:

$$x_{ijt} \equiv \log\left(X_{ijt}\right) - \frac{\sum_{k:k\in\text{i's industry in year t}} \log X_{kjt}}{\|\{k : k \in \text{i's industry in year t}\}\|} \tag{23}$$

4 Implications of Materials Price Dispersion

In this section, I explore some of the implications of price dispersion in intermediate inputs markets. Correlations among plant-level statistics are given in Section 4.1. In Section 4.2, I estimate that 7 to 16 per cent of the variation in $tfpq$ is attributable to differences in the materials prices that plants face. In Section 4.3, I argue that the price that plants face when purchasing their materials is persistent across time. In Section 4.4, I document that materials prices are substantially higher for exiting plants (compared to survivors) and neither higher nor lower for entrants (compared to incumbents). Then, in Section 4.5, I compute the contribution, towards aggregate productivity growth, of the entry of relatively productive plants and the exit of relatively unproductive plants.

[33] Of course, there may be within-industry differences in the factor market conditions for labor, capital, and electricity. Because of Assumption 2, these differences would be incorrectly labeled as differences in "technical efficiencies."

4.1 Correlations

I begin by computing the correlations among different plant-level statistics. All plant-level variables are de-meaned by industry-year according to Equation 23. Correlations are weighted by plants' real sales.[34] The correlations, which are collected in Table 2, match the predictions made in parts 1-6 of Proposition 2.

The correlation coefficients between $tfpq$, $tfpr$, and p^{out} are similar to those computed in Foster, Haltiwanger, and Syverson (2008). Plants with higher $tfpq$ have a lower marginal cost and pass on some of this lower marginal cost to their consumers (generating a low p^{out}). In addition, $tfpq$ and $tfpr$ are highly correlated, as are $tfpr$ and p^{out}.

The variables that are new to this study are ϕ and p^{in}, the logged technical efficiencies and materials prices. First, plant-level materials prices, p^{in}, are negatively correlated with $tfpq$. Plants that purchase inputs cheaply appear to be more productive. At the same time, $tfpq$ and ϕ are highly correlated with one another, while the correlation between ϕ and $tfpr$ is similar to the correlation between $tfpr$ and $tfpq$. These last correlations should bring some solace to researchers who do not have access to plant-level materials and output prices, and who wish to use $tfpr$ as a proxy for ϕ.

Finally, plant-level input and output prices are correlated with one another.[35] There are several possible explanations for this positive relationship. First, the positive correlation may reflect that some differences in input and output quality still remain (despite my best efforts to choose a sample of industries with outputs and material inputs that are comparable across plants). Independent of quality differences, the positive correlation between input and output prices may be due to imperfections in output markets, as Proposition 2 would predict. A third possible explanation is that a selection mechanism, one on plant survival, may be causing us to observe a positive relationship between p^{out} and p^{in} (see Equations 8 and 9).

Correlations for each of the ten industries are presented in Appendix D.5. It seems that, within the benchmark sample, the sugar industry is something of an outlier: for the 177 plant-year observations in the industry, the correlation between p^{in} and $tfpq$ is -3%, not significantly different from 0. Except for sugar, the range of correlations between p^{in} and $tfpq$ lies between -28% (packaged milk) and -49% (coffee).

[34] Throughout this section, real revenues are computed using 4-digit-industry output price indices from the NBER Productivity Database. Unweighted calculations are presented in Appendix D.6.

[35] Eslava et al. (2004) document this empirical regularity, as well, using a broad sample of Colombian manufacturing establishments.

	p^{in}	p^{out}	$tfpq$	ϕ	$tfpr$
p^{out}	0.231*				
$tfpq$	-0.369*	-0.551*			
ϕ	0.127*	-0.469*	0.873*		
$tfpr$	-0.232*	0.219*	0.616*	0.694*	
Std. Dev.	0.117	0.119	0.161	0.151	0.137
Skewness	-0.135	0.127	0.739	0.669	1.201

Table 2: Correlations and sample statistics of plant-level characteristics, pooled across all products. All variables are de-meaned by year and product. N=10,503. Observations are weighted by plants' real revenues. Stars indicate that the correlation is significantly different from 0, at the 5% level (see Appendix E for details). Correlations that give plant-year observations equal weight are presented in Appendix D.6.

4.2 Implications for productivity dispersion

In this subsection, I compare the dispersions of the distributions of $tfpq$ and ϕ. In so doing, I provide a measure of the fraction of $tfpq$ dispersion that can be explained by differences in intermediate input prices. The main finding, that dispersion in $tfpq$ is significantly greater than the dispersion of ϕ, is the prediction of part 7 of Proposition 2.

Pooling across the 10 industries in the sample, the standard deviation of ϕ is 16.3% ($= e^{0.151}$), while the standard deviation of $tfpq$ is 17.5%, 7% larger than the standard deviation of ϕ. So, by purging out the effect of differences in materials prices, the observed distribution of productivities would be approximately 7% lower; the 95% confidence interval of the difference between the standard deviations of $tfpq$ and ϕ is $[0.2\%, 10.4\%]$.

I also give, in Table 3, two other measures of dispersion, the 90/10 ratio and the 75/25 ratio. The difference between the dispersion of $tfpq$ and the dispersion of ϕ is even greater with these two alternate measures: 9% for the 90/10 ratio, and 10% for the 75/25 ratio.

Again, even though I have chosen industries based on the homogeneity of the inputs and outputs, it is likely that at least some of the variation in materials and output prices is due to differences in quality. Variation in input/output quality attenuates the negative correlation between $tfpq$ and p^{in} (see Appendices D.1 and D.2.) As a result, within-industry variation in quality will lead to a downward bias in the measured difference between the dispersion of $tfpq$ and the dispersion of ϕ. In other words, the 7% to 10% decline in dispersion most likely underrepresents the actual fraction of $tfpq$ dispersion that is due to differences in materials prices.

To provide some context, I relate the 7% to 10% decline in dispersion to dispersion declines reported in two other papers. First, Syverson (2004a) hypothesizes that, in markets

	Dispersion of $tfpq$			Dispersion of ϕ			Percent Decline		
Sample	90/10	75/25	SD	90/10	75/25	SD	90/10	75/25	SD
Milk-Bulk	0.597	0.285	0.252	0.531	0.267	0.229	13.1%	7.2%	10.3%
Flour	0.360	0.190	0.142	0.349	0.158	0.148	3.2%	23.2%*	-4.1%
Concrete	0.521	0.251	0.224	0.486	0.236	0.215	7.4%*	6.6%*	4.4%*
Yarn	0.581	0.275	0.256	0.633	0.310	0.252	-8.0%	-10.8%	1.5%
Boxes, Yr.$\leq$'87	0.380	0.179	0.168	0.366	0.177	0.166	3.7%*	1.0%	1.6%
Gasoline	0.300	0.145	0.132	0.280	0.133	0.122	7.6%	9.5%	8.1%
Coffee	0.709	0.326	0.266	0.562	0.277	0.229	30.1%*	19.5%*	17.4%*
Sugar	0.588	0.297	0.280	0.766	0.340	0.319	-20.7%	-11.9%	-11.5%*
Boxes, Yr.$\geq$'92	0.566	0.293	0.225	0.526	0.276	0.211	7.9%*	6.2%*	6.7%*
Milk-Packaged	0.535	0.261	0.227	0.502	0.248	0.218	6.7%*	5.3%*	4.1%*
Pooled: Weighted	0.351	0.164	0.161	0.324	0.151	0.151	8.8%*	9.7%	6.9%*
Pooled: Unweighted	0.527	0.253	0.227	0.493	0.238	0.219	7.2%*	6.5%*	3.9%*

Table 3: Dispersion of $tfpq$ and ϕ. In the final three columns, stars indicate that the difference between $tfpq$ and ϕ is statistically significant, at the 5% level (see Appendix E for details). Except for the final row, observations are weighted by plants' real revenues. See Table 28 in Appendix D.6 for the unweighted computations, broken out by industry. Due to Census' rules regarding data confidentiality, I am prohibited from reporting the actual quantiles of any empirical distribution. The quantiles (but not the standard deviations) are computed in a two-step process. First, using a kernel density estimator, I produce a smoothed version of the empirical cumulative distribution function of the variable of interest. I then report the quantile of this smoothed distribution. The decrease in productivity dispersion—between $tfpq$ and ϕ—is not substantially affected by this smoothing procedure.

Revenue-	Dispersion of $tfpr$			Dispersion of $tfpq$			Percent Increase		
weighted?	90/10	75/25	SD	90/10	75/25	SD	90/10	75/25	SD
Yes	0.306	0.147	0.137	0.351	0.164	0.161	16.0%*	12.9%	18.5%*
No	0.380	0.178	0.176	0.527	0.253	0.227	47.3%*	53.0%*	33.7%*

Table 4: Dispersion of $tfpr$ and $tfpq$. In the final three columns, stars indicate that the difference between $tfpr$ and $tfpq$ is statistically significant, at the 5% level (see Appendix E for details). Due to Census' rules regarding data confidentiality, I am prohibited from reporting the actual quantiles of any empirical distribution. See the caption of Table 3 for a description of the imputation of the quantiles of $tfpq$ and $tfpr$. N=10,503.

for which competitive forces are exceptionally strong, low-productivity plants are relatively more apt to exit the industry, thus leading to a more compressed productivity distribution. Within the ready-mix concrete industry, Syverson characterizes areas with high densities of construction activity as highly competitive markets. Compared to low-density markets, the interquartile range of total factor productivity is 2 to 3% smaller in high-density markets. In a second example, Fox and Smeets (2011) compute the fraction of measured productivity dispersion that can be explained by differences in worker quality. For a sample of Danish firms in eight industries, the 90/10 ratio of the distribution of TFP declines by approximately 20% after including controls for worker quality and the wage bill.

While price dispersion in intermediate input markets tends to reduce the dispersion of measured productivity (i.e., the dispersion of $tfpq$ is greater than that of ϕ), price dispersion in output markets has the opposite effect on the dispersion of measured productivity (i.e., the dispersion $tfpr$ is smaller than that of $tfpq$). The latter relationship, which Foster, Haltiwanger, and Syverson (2008) also document, stems from the strong negative correlation between p^{out} and $tfpq$: The standard deviation of revenue productivity, which is 14.7% in the revenue-weighted calculations, is approximately 19% smaller than the standard deviation of quantity productivity. The countervailing effects–as in this case, on the standard deviation of measured productivity–of factor price dispersion and goods price dispersion will be a recurring finding in the remainder of this section.

4.3 Persistence

A long stream of research, beginning with Baily, Hulten, and Campbell (1992), has documented the persistence of plant-level characteristics. Using regressions of the form,

$$x_{i,j,t+5} = \alpha + \beta \cdot x_{ijt} \ + \varepsilon_{ijt}, \tag{24}$$

Foster, Haltiwanger, and Syverson (2008) compute the 1 and 5-year autocorrelation co-

	Revenue-weighted?	$tfpq$	ϕ	$tfpr$	y	p^{in}	p^{out}
β	No	0.351*	0.299*	0.306*	0.895*	0.309*	0.432*
$s.e.$	No	(0.022)	(0.023)	(0.024)	(0.010)	(0.025)	(0.024)
$\beta^{1/5}$	No	0.811	0.786	0.789	0.978	0.791	0.846
β	Yes	0.175	0.185*	0.201	0.868*	0.326*	0.305*
$s.e.$	Yes	(0.092)	(0.083)	(0.106)	(0.045)	(0.047)	(0.040)
$\beta^{1/5}$	Yes	0.706	0.713	0.726	0.972	0.799	0.789

Table 5: Coefficient estimates and robust standard errors of the regression defined by Equation 24. Stars indicate significance at the 5% level. N=4310.

efficients for different plant-level statistics, x. They compute that plant-level productivities and output prices have a 1-year autocorrelation coefficient of approximately 80%. I replicate these findings in Table 5. The novel components of Table 5 appear in the second, fourth, and penultimate columns, where I compute the autocorrelation coefficients of ϕ, y, and p^{in}. I find that the persistence of ϕ is similar to that of the two other plant-level productivity measures, and that the persistence of p^{in} not statistically different from the persistence of p^{out}. By far, the most persistent plant-level characteristic is log revenues, y: the 1-year autocorrelation coefficient is approximately 97%.

4.4 Characteristics of entering and exiting plants

Along many important dimensions, plants that are about to exit are different from surviving plants, while plants that are entering an industry are different from incumbents. Several studies have documented a negative relationship between productivity (or size) and probability of survival. Moreover, while entrants are smaller than the average plant in their industry, they are not less productive.

I replicate these results in Table 6, using regressions defined by Equations 25 and 26. Like Foster, Haltiwanger, and Syverson (2008), I find that entrants have above-average $tfpq$ and $tfpr$, but below-average revenues. These authors also estimate that, compared to other plants in the industry-year, exiting plants have lower $tfpq$ and $tfpr$. The productivity advantage of entrants, and productivity disadvantage of exiting plants, is larger for quantity productivity than it is for revenue productivity: removing the output-price component of revenue productivity tends to decrease the difference between surviving and exiting plants' productivitties.

$$x_{ijt} = \alpha_{jt} + \beta_1 \mathbb{I}\{i \in \text{Entrant at year } t\} + \varepsilon_{ijt} \quad (25)$$

$$x_{ijt} = \alpha_{jt} + \beta_2 \mathbb{I}\{i \in \text{Exit before } t+5\} + \varepsilon_{ijt} \quad (26)$$

Coefficient on:	Revenue-weighted?	$tfpq$	ϕ	$tfpr$	y	p^{in}	p^{out}
Entry	No	0.017*	0.020*	0.015*	-0.456*	0.005	-0.002
Entry	No	(0.008)	(0.007)	(0.006)	(0.032)	(0.006)	(0.006)
Entry	Yes	0.009	-0.004	-0.005	-0.698*	-0.019	-0.014
Entry	Yes	(0.025)	0.023)	(0.022)	(0.092)	(0.015)	(0.016)
Exit	No	-0.025*	-0.016*	-0.020*	-0.534*	0.014*	0.005
Exit	No	(0.007)	(0.007)	(0.006)	(0.031)	(0.005)	(0.006)
Exit	Yes	-0.051*	-0.042*	-0.039	-0.609*	0.013	0.012
Exit	Yes	(0.018)	(0.016)	(0.020)	(0.131)	(0.012)	(0.012)

Table 6: Comparison of plant-level statistics and entry/exit status. In the first four rows, each cell gives the coefficient estimate, or standard error, of β_1 in Equation 25. In the final four rows, each cell gives the coefficient estiamte, or standard error, of β_2 in Equation 26. Stars denote significance at the 5% level. N=10,503.

In addition to these already-known empirical regularities, I compute, in the second-to-last column of Table 6, that exiting plants pay, approximately 1.4% (1.3% for the revenue-weighted calculations) more per unit of the intermediate input than the average plant in their industry-year. When observations are weighted equally, the technical efficiency, ϕ, is 2% higher for entrants (compared to incumbents), and 1.6% lower for exiting plants (compared to survivors). Comparing the first two columns of 6, the productivity advantage of incumbents is larger for $tfpq$ than it is for ϕ: Removing the materials-price component from quantity productivity decreases the measured difference between surviving and exiting plants' productivitties. In summation, Table 6 states that materials price variation and output price variation have opposite effects on the measured productivity disadvantage of exiting plants.

Interactions among different selection mechanisms are explored in Table 7. This table presents the results of a logit regression, in which the dependent variable equals 1 if the plant exits before the subsequent Census of Manufacturers. Besides industry-year fixed effects, the regressions include characteristics that affect plants' probabilities of survival. The effects of increasing measured productivity or materials prices are considered, in isolation, in the first four columns of Table 7. A one standard deviation increase in $tfpq$, $tfpr$, and ϕ is associated with a 1.2%, 1.2%, or 0.8% decline in the probability of exit, while a one standard deviation increase in p^{in} is associated with a 1.1% increase in the probability of exit. Including plant-level input prices, as I do in the fifth and sixth columns, attenuates the estimates of the $tfpr$ and $tfpq$ terms, since some component of the selection-on-productivity mechanism is now being captured by the p^{in} variable. Note that no such attenuation occurs between the third/fourth and seventh columns.

Tables 6 and 7 indicate that part of the selection-on-profitability mechanism is due

$tfpr$	-0.594*				-0.552*		
	(0.179)				(0.180)		
$tfpq$		-0.458*				-0.380*	
		(0.133)				(0.139)	
ϕ			-0.315*				-0.369*
			(0.138)				(0.139)
p^{in}				0.530*	0.472*	0.375*	0.594*
				(0.180)	(0.182)	(0.189)	(0.181)

Table 7: Coefficient estimates and standard errors (clustered at the plant level) from a logit regression: the depedent variable equals 1 provided the plant exits before the following Census of Manufacturers. Stars denote significance at the 5% level. N=10,503.

to the higher-than-average intermediate input prices that exiting plants pay. Relatedly, the productivity advantage of entrants (compared to incumbents) and surviving plants (compared to exiting plants) is highest when using $tfpq$ as the productivity measure. In other words, controlling for output prices, but not materials prices, tends to make entrants (survivors) appear relatively more productive, compared to incumbents (exiting plants). The next subsection considers the magnitude and significance of the differences, across the three productivity measures, of the importance of reallocation via entry.

4.5 Decompositions of industry productivity growth

In this subsection, I compute the fraction of aggregate productivity growth that occurs via the *net entry effect:* the exit of relatively unproductive plants and the entry of relatively productive plants. The extent to which reallocation across plants explains aggregate productivity growth has been extensively studied (e.g., Baily, Hulten, and Campbell (1992), Griliches and Regev (1995), Foster, Haltiwanger, and Krizan (2001), and Foster, Haltiwanger, and Syverson (2008)). Of these analyses, I am following, most closely, Section 5.2 of Foster, Haltiwanger, and Syverson (2008). There, the authors compute the net entry effect when either $tfpr$ or $tfpq$ is used as the measure of plant productivity. Because entrants charge significantly lower prices, compared to incumbents, the net entry effect is smaller when revenue productivity measures are used instead of quantity productivity measures. The authors conclude that, "in terms of understanding the barriers to allocative efficiency ... revenue based productivity decompositions may focus too much attention on continuing businesses and not enough on the role of entering businesses." (page 419). Below, I show that accounting for materials prices substantially reverses this finding. The net entry effect is larger when $tfpq$, instead of ϕ, is used as the measure of plant productivity.

Like Foster, Haltiwanger, and Syverson (2008), I use the following two growth de-

compositions, the first due to Baily, Hulten, and Campbell (1992) and Foster, Haltiwanger, and Krizan (2001) and the second due to Griliches and Regev (1995):

$$\begin{aligned}\Delta\overline{tfp}_t &= \sum_{i\in\mathcal{C}}\theta_{i,t-1}\Delta tfp_{it} + \sum_{i\in\mathcal{C}}\left(tfp_{i,t-1}-\overline{tfp}_{t-1}\right)\Delta\sigma_{it} + \sum_{i\in\mathcal{C}}\Delta tfp_{it}\Delta\theta_{it} \\ &\quad \underbrace{+\sum_{i\in\mathcal{N}}\theta_{i,t-1}(tfp_{it}-\overline{tfp}_{t-1}) - \sum_{i\in\mathcal{X}}\theta_{i,t-1}(tfp_{i,t-1}-\overline{tfp}_{t-1})}_{\text{Entry Effect}\qquad\qquad\text{Exit Effect}}\end{aligned} \tag{27}$$

$$\begin{aligned}\Delta\overline{tfp}_t &= \sum_{i\in\mathcal{C}}\frac{1}{2}\left(\theta_{i,t-1}+\theta_{i,t}\right)\Delta tfp_{it} + \sum_{i\in\mathcal{C}}\frac{1}{2}\left(tfp_{i,t}+tfp_{i,t-1}-\overline{tfp}_{t-1}-\overline{tfp}_t\right)\Delta\theta_{it} \\ &\quad \underbrace{+\sum_{i\in\mathcal{N}}\theta_{it}\left(tfp_{it}-\frac{1}{2}\overline{tfp}_{t-1}-\frac{1}{2}\overline{tfp}_t\right) - \sum_{i\in\mathcal{X}}\theta_{i,t-1}\left(tfp_{i,t-1}-\frac{1}{2}\overline{tfp}_{t-1}-\frac{1}{2}\overline{tfp}_t\right)}_{\text{Entry Effect}\qquad\qquad\text{Exit Effect}}\end{aligned} \tag{28}$$

In Equations 27 and 28, θ_{it} denotes the revenue share of plant i, within its industry, in year t; $\overline{tfp}_t$ gives the revenue-weighted average (log) productivity in year t; Δ is the difference operator; and $\mathcal{C}$, $\mathcal{N}$, and $\mathcal{X}$ are the sets of continuing, entering, and exiting plants. The decompositions highlight the different sources of industry productivity growth, including the the Entry Effect, the Exit Effect, and the sum of the two effects (the Net Entry Effect).[36] The magnitudes of these three effects will depend on which productivity measure–either $tfpr$, $tfpq$, or ϕ–that is being plugged into Equations 27-28.

The results of the industry decompositions are given in Table 8. I decompose the productivity growth–over 5 year intervals–separately for each of the 10 industries in the benchmark sample. The values reported in the table are the averages over these 10 industries. In the first 3 rows, industries with larger sales (primarily gasoline manufacturing) are given more weight, while, in the last 3 rows, industries' weights are determined by the number of plants in the industry. The main takeaway from the table is that the Net Entry term is larger for quantity productivity ($tfpq$) than it is for either revenue productivity ($tfpr$) or technical efficiency (ϕ). So, consistent with Foster, Haltiwanger, and Syverson (2008), Table 8 indicates that the contribution of net entry to aggregate productivity is larger when output prices are accounted for. At the same time, accounting for materials prices reduces the measured contribution of net entry to productivity growth. These patterns are robust to the decomposition method and the relative weights given to different industries.[37]

[36] Since a large number of plants enter and exit my benchmark sample without actually entering or exiting their industries, I will be unable to distinguish among the sources of aggregate productivity growth that are listed in the first lines of Equations 27 and 28.

[37] One problem with the productivity growth decompositions originates from the overrepresentation, in

Productivity Measure	Weight Industries By:	Total	Entry	Exit	Net Entry	Entry	Exit	Net Entry
$tfpr$	Real Revenues	-1.60	-0.08	0.08	0.00	-0.05	0.12	0.08
$tfpq$	Real Revenues	-1.60	-0.06	0.15	0.09	-0.02	0.19	0.16
ϕ	Real Revenues	-1.60	-0.09	0.13	0.04	-0.05	0.17	0.12
$tfpr$	# of Plants	1.30	0.23	0.12	0.35	0.16	0.22	0.38
$tfpq$	# of Plants	1.30	0.32	0.12	0.44	0.25	0.22	0.47
ϕ	# of Plants	1.30	0.25*	0.14	0.39	0.17*	0.24	0.41
Decomposition Method			Foster et. al			Griliches and Regev		

Table 8: Aggregate productivity growth decompositions (all values are given as percentages). Stars indicate that the value given in the cell is significantly different than the corresponding value that uses $tfpq$ as the measure of plant productivity. See Appendix E for details.

For completeness' sake, I assess the statistical significance of the differences, across the productivity measures, of the importance of the Entry, Exit, or Net Entry terms. When industries are weighted by the number of plants, the Entry Effect is significantly greater when ϕ–instead of $tfpq$–is used as the productivity measure. Other differences are not statistically significant.

5 Sources of Materials Price Dispersion

I discuss three explanations for the observed within-industry dispersion of intermediate input prices. The sources of materials price dispersion have implications for the social benefits generated by each plant. Plants that pay low materials prices by taking advantage of monopsonistic power are not providing any societal benefit: low materials prices are a transfer of profits from supplier to buyer. On the other hand, if plants pay low materials prices because their suppliers are exceptionally productive, low materials prices represent a positive impact on social welfare.

To calculate the relative importance of these different sources of materials price dispersion, I need to impute, for each manufacturer, the identities of its suppliers. I outline, in Section 5.1, the algorithm that I use to impute buyer-supplier relationships. In Section 5.2, I compute the fraction of dispersion in $tfpq$ and p^{in} that can be explained by plants' geographic locations and their suppliers' marginal costs. A positive correlation between

the benchmark sample, of large plants. Because of this, entering and exiting plants are underrepresented in the benchmark sample. As a result, the decompositions understate the role of net entry as a source of aggregate productivity growth.

In Appendix D.7, on page 61, I show that the qualitative patterns of this section (in particular, the difference, between the three productivity measures, in the size of the Net Entry Effect) hold after correcting for the underrepresentation of entering and exiting plants.

plants' marginal costs and their suppliers' marginal costs stimulates the following question: If plants with low-marginal cost suppliers are more productive, and having low marginal costs is so advantageous, what prevents plants from purchasing their materials inputs from the low marginal cost suppliers? In Section 5.3, I argue that buyer-supplier relationships are persistent, suggesting that there is some inertial force that inhibits all plants from switching to low marginal cost suppliers. I close, in Section 5.4, by showing that suppliers charge different prices to different customers, *prima facie* evidence of price discriminatory behavior. I calculate the fraction of materials price dispersion that can be explained by this price discriminatory behavior.

5.1 Imputation of buyer-supplier relationships

To impute buyer-supplier relationships, I use the algorithm introduced in Atalay, Hortaçsu, and Syverson (2012). The algorithm generates a list of establishments that could potentially receive any shipment that is observed in the Commodity Flow Survey. Consider a hypothetical shipment of commodity, c, made by establishment, h, to zip code, z.[38] The establishments, i, that could potentially receive this shipment are those who are located in zip code, z, and are members of an industry that use commodity, c. For example, the potential recipients of a shipment of Portland cement to z would be all plants in that zip code who were engaged in road construction, concrete brick manufacturing, ready-mix concrete manufacturing, or wholeasaling of brick, stone and related materials. If there are multiple potential recipients of the shipment, and one of these establishments is owned by the same firm as the sending establishment, then I presume that the shipment was received by the same-firm establishment.[39] Otherwise, I assign each potential recipient, i, to be downstream of plant h.[40]

So that I can compare suppliers' and buyers' productivities, I require both the upstream and downstream industries to be from the manufacturing sector. Of the ten industries in the benchmark sample, only two–ready-mix concrete and cardboard boxes–have a

[38] The commodity code used in the 1993 Commodity Flow Survey is the Standard Transportation Commodity Code (STCC). A list of STCC codes can be found in pages 117 to 167 of "Reference Guide for the 2008 Surface Transportation Board Carload Waybill Sample," published by Railinc. Since 1997, the Commodity Flow Survey has used the Standard Classification of Transported Goods (SCTG) classification of commodity codes. Documentation related to SCTG codes can be found on the Census website.

[39] Atalay, Hortaçsu, and Syverson (2012) make the same assumption. This assumption is motivated by the finding that establishment h is much more likely to ship to zip codes that contain an establishment from the same firm.

[40] By assigning all potential recipients, i, to be downstream of plant h, I am undoubtedly overcounting the number of buyer-supplier relationships. In an unreported robustness check, I reproduce the analysis of Section 5.2, weighting observations by the inverse of the number of potential recipients in the destination zip code. I find that the results of Section 5 are essentially unchanged.

main input that is produced by a manufacturer. For Portland cement producers, I look for shipments in the Commodity Flow Survey for which the commodity code is that of cement.[41] The industries with establishments that could potentially receive Portland cement (STCC=32411) are road construction firms (SIC=1610-1619), concrete brick and block manufacturers (SIC=3271), ready-mix concrete manufacturers (SIC=3273), and wholesalers of brick, stone and related materials (SIC=5032). For paper and paperboard manufacturers, I look for shipments in the Commodity Flow Survey for which the commodity code is that of paperboard (STCC=26311 in 1993, SCTG=27319-27320 in 1997). The industries with establishments that could potentially receive these shipments of paperboard are manufacturers of corrugated and solid fiber boxes (SIC=2653) and folding paperboard boxes (SIC=2657).

For within-firm shipments, surveyed establishments do not report the actual market value of the transaction. Instead, the establishments are asked to estimate what the value of the shipment would have been if it had been sold to some other firm. Since it is unclear what these values actually represent, I remove downstream establishments who receive a substantial fraction, 15% or more, of the relevant input from other plants from the same firm.[42]

Mismeasurement of buyer-supplier relationships will primarily affect the estimated strength of the relationship between buyers' and suppliers' productivities. In other exercises, I a) compare downstream plants' materials prices and upstream plants' productivities, and b) examine the persistence of buyer-supplier relationships. These exercises rely only on information about the sending establishments, and thus are not as sensitive to mismeasurement of buyer-supplier relationships.

5.2 Materials prices and supplier marginal costs.

I document the following two relationships. First, ready-mix concrete and cardboard box manufacturers that have exceptionally productive suppliers purchase their cement or paperboard at lower than average prices. This finding, along with the negative relationship between a plant's input prices and its $tfpq$ (which is documented in Section 4.2), combines

[41] Productivity data for ready-mix concrete manufacturers are unavailable in 1997. So, for cement and concrete manufacturers, I only look at buyer-supplier relationships in the 1993 Commodity Flow Survey.

[42] While varying the 15% cutoff down to 0% or up to 25% does not affect this section's results, the relationship between input prices and supplier productivity begins to disappear once the cutoff exceeds 25 or 30%.

Bernard, Jensen, and Schott (2006) show that reported prices on cross-border shipments, for which the sender and receiver are part of the same firm, are manipulated to take advantage of the different tax policies of the destination and source countries. Even though such an incentive to mis-report does not exist in the Commodity Flow Survey data, I argue that one should not put too much weight on input prices of the plants that buy a substantial fraction of their inputs from within the firm.

to generate the prediction of a second relationship: the $tfpq$ of a plant and the $tfpq$ of its suppliers should be positively correlated. These relationships match the predictions contained in Proposition 3.

I begin with some notation. Let χ_{hit} denote the total mass (in thousands of pounds) of shipments sent by plant h to plant i in year t, and let ω_{hit} denote the total value (in thousands of real dollars) of shipments sent by plant h to plant i in year t. Then, the (f.o.b.)[43] price that plant h charges plant i is simply the ratio of the value to the price:

$$P_{hit}^{CFS} = \frac{\omega_{hit}}{\chi_{hit}} \tag{29}$$

The "CFS" superscript will denote prices computed using the Commodity Flow Survey data (as opposed to the prices that are computed in Section 4, using data from the Census of Manufacturers.)[44]

A supplier's average output price equals a value-weighted average of the prices that it charges to the plants that we observe it supplying to:

$$P_{ht}^{out,CFS} = \frac{\sum_{i:h\to i} \omega_{hit} P_{hit}^{CFS}}{\sum_{i:h\to i} \omega_{hit}} \tag{30}$$

Plants' average input prices are defined similarly. For each downstream plant, i, I take the value-weighted average over all plants, h, that I observe i purchasing from:

$$P_{it}^{in,CFS} = \frac{\sum_{h:h\to i} \omega_{hit} P_{hit}^{CFS}}{\sum_{h:h\to i} \omega_{hit}} \tag{31}$$

Note that, because it does not include freight charges, $P_{it}^{in,CFS}$ will be less than what plants pay for their intermediate inputs. I define a second plant-level input price, that includes freight charges:

$$\tilde{P}_{it}^{in,CFS} = \frac{\sum_{h:h\to i} \omega_{hit} \left(P_{hit}^{CFS} + \tau_{hit}\right)}{\sum_{h:h\to i} \omega_{hit}} \tag{32}$$

I estimate transportation costs, τ_{hi}, from the mileage of the shipment and the mode of

[43] F.o.b. stands for free on board (or freight on board). Unlike the c.i.f. (cost, insurance, and freight) price, the f.o.b. price does not include freight or insurance charges.

[44] The Commodity Flow Survey has, up to now, been an unexploited source of data on plants' output prices. With this in mind, I compare plants output prices, derived from the Commodity Flow Survey to the prices derived from the better-known Census of Manufacturers. For the 53 cement manufacturers in this section's sample, the correlation between $p_h^{out,CFS}$ and p_h^{out} is 21%. For the 60 paperboard manufacturers in the sample, the correlation between the two plant-level output prices is 41%.

transport.[45,46]

Finally, for any concrete or cardboard box manufacturer, i, that is identified by the algorithm outlined in Section 5.1, I compute average supplier productivity, $\overline{TFPQ}_{it}$ as follows:

$$\overline{TFPQ}_{it} \equiv \frac{\sum_{h:h\to i} \omega_{hit} TFPQ_{ht}}{\sum_{h:h\to i} \omega_{hit}} \tag{33}$$

Similar to the analysis of Section 4, all plant-level statistics are logged and stated relative to the average value for the industry-year. Again, the logged, de-meaned variables are written using lower-case letters.

With these definitions in hand, I am now able to compare the price that a plant pays for its material inputs to the average productivity of the plant's suppliers.[47]

$$\tilde{p}_{it}^{in,CFS} = \beta_{\text{division}} + \beta_1 \cdot \overline{tfpq}_{it} + \varepsilon_{it} \tag{34}$$

The results are presented in Table 9. A 10 percent increase in the marginal cost of plants' suppliers corresponds to a 2 to 2.5 percent increase in plants' materials prices. The estimated effect of supplier productivity on materials prices is somewhat stronger for boxes than it is for ready-mix concrete. Including fixed effects for the geographic region of the downstream plant slightly decreases the estimate of β_1: one reason why intermediate input prices are spatially correlated is that the plants within a particular region are likely to have the same suppliers. (An alternate explanation is that marginal costs of plants in the supplying industry are spatially correlated.)

Each cell in Table 10 presents the unexplained variation–measured as the (revenue-weighted) standard deviation of the residuals, when $\tilde{p}_{it}^{in,CFS}$ is regressed on different combinations of the right-hand side variables of Equation 34. Comparing the first and second

[45] The Bureau of Transportation Statistics collects information on ton-mile freight charges for shipments sent along different transport modes. Since the Commodity Flow Survey contains information on the weight of each shipment, as well as the distance that the shipment traveled, it is straightforward to estimate the shipment freight charge.

See http://www.bts.gov/publications/national_transportation_statistics/html/table_03_21.html for the data on freight rates.

[46] For the corrugated-box manufacturing industry, I relate $\tilde{p}_{it}^{in,CFS}$ and p_{it}^{in}. (Remember that p_{it}^{in} cannot be computed in 1992 or 1997 for ready-mix concrete manufacturers.) The strength of this relationship, between the materials prices computed from the two data sources, indicates the success (or lack thereof) of the imputation procedure outlined in Section 5.1. The correlation between $\tilde{p}_{it}^{in,CFS}$ and p_{it}^{in} is 25%, meaning that I am mismeasuring many buyer-supplier relationships, but that the imputation algorithm yields a workable dataset.

[47] Using $p_{it}^{in,CFS}$, instead of $\tilde{p}_{it}^{in,CFS}$, as the dependent variable of Regression 34 generates a similar estimate of β_1.

Sample	Boxes	Concrete	Pooled	Boxes	Concrete	Pooled
$\overline{tfpq_{it}}$	-0.261*	-0.189*	-0.245*	-0.253*	-0.188	-0.237*
	(0.059)	(0.093)	(0.050)	(0.058)	(0.112)	(0.049)
N	190	131	321	190	131	321
Adjusted R^2	0.104	0.036	0.087	0.099	0.084	0.091
Division F.E.?	No	No	No	Yes	Yes	Yes

Table 9: Coefficient estimates and robust standard errors, from the regressions defined by Equation 34. Stars indicate significance at the 5% level.

Sample	Include Division Fixed Effects ? Include $\overline{tfpq_{it}}$? Sample Size	No No	Yes No	No Yes	Yes Yes
Boxes	190	0.202	0.197	0.190*	0.187
Concrete	131	0.374	0.351	0.365	0.345
Pooled	321	0.224	0.220	0.214*	0.211*

Table 10: Each cell gives the real-revenue-weighted standard deviation of the residuals in a particular regression; the full specification is given in Equation 34. Across the columns of the table, different combinations of independent variables are included in the regressions. Across the rows of the table, different samples are used. Stars indicate that the decline in dispersion is significantly more than the decline that would occur from simply including "fake" random variables on the right hand side of Equation 34. See Appendix E for details.

columns of Table 10, I calculate that the inclusion of division fixed effects reduces the unexplained variation of materials prices by approximately 1.8% $\left(= \frac{0.224-0.220}{0.224}\right)$. The inclusion of suppliers' productivities reduces the unexplained variation by approximately 4.4% $\left(= \frac{0.224-0.214}{0.224}\right)$, while the two sets of variables jointly reduce the unexplained price variation by 5.8%. Stars in Table 10 indicate that the decline in dispersion is significantly more than what would be expected by including randomly-generated regressors.[48]

A prediction that emerges from two previously documented relationships–the negative relationship between p^{out} and $tfpq$, and the negative relationship between $\tilde{p}_{it}^{in,CFS}$ and $\overline{tfpq_{it}}$–is that plant level $tfpq$ should be positively correlated to the $tfpq$ of the plants' suppliers. I test this prediction by running the regression specified by Equation 35.

$$tfpq_{it} = \beta_{\text{division}} + \beta_1 \cdot \overline{tfpq_{it}} + \varepsilon_{it} \tag{35}$$

[48] Any set of variables–for example, a random variable drawn from a standard normal distribution, or a set of 12 dummy variables that sum up to 1–will necessarily explain some positive fraction of the variation in $\tilde{p}_i^{in,CFS}$. In Appendix E, on page 65, I test whether the decline in dispersion is significantly greater than what would be expected from including different combinations of "fake" random variables on the right hand side of Equation 34.

Sample	Boxes	Concrete	Pooled	Boxes	Concrete	Pooled
$\overline{tfpq_{it}}$	0.078	0.132*	0.090	0.027	0.133*	0.070*
	(0.066)	(0.045)	(0.052)	(0.063)	(0.054)	(0.047)
N	190	131	321	190	131	321
Adjusted R^2	0.003	0.059	0.010	0.034	0.052	0.042
Division F.E.?	No	No	No	Yes	Yes	Yes

Table 11: Coefficient estimates and robust standard errors of the regressions defined by Equation 35. Stars indicate significance at the 5% level.

	Include Division Fixed Effects ?	No	Yes	No	Yes
	Include $\overline{tfpq_{it}}$?	No	No	Yes	Yes
Sample	Sample Size				
Boxes	190	0.219	0.211	0.218*	0.210
Concrete	131	0.210	0.203	0.203*	0.197
Pooled	321	0.218	0.211	0.217	0.211

Table 12: Each cell gives the real-revenue-weighted standard deviation of the residuals in a particular regression; the full specification is given in Equation 35. Across the columns of the table, different combinations of independent variables are included in the regressions. Across the rows of the table, different samples are used. Stars indicate that the decline in dispersion is significantly more than the decline that would occur from simply including "fake" random variables on the right hand side of Equation 35. See Appendix E for details.

The results of this regression, which are displayed in Table 11, confirm that there is, indeed, a positive relationship between plants' marginal costs and the marginal costs of plants' suppliers. The relationship between $tfpq_{it}$ and $\overline{tfpq_{it}}$ is weaker, however, than the relationship between $tfpq_i$ and $\tilde{p}_{it}^{in,CFS}$ that was estimated in Table 9.

Table 12 shows the unexplained variation of $tfpq_{it}$ that remains after regressing on different combinations of explanatory variables. Only 4% of the variation in $tfpq_{it}$ can be accounted for, using division fixed effects and suppliers marginal costs.

5.3 Persistence of relationships

Buyer-supplier relationships are persistent across time. The estimated persistence of buyer-supplier relationships suggests that there is some force that inhibits intermediate inputs purchasers from switching suppliers. Whether this inhibiting force reflects some extra profitability that is conferred by repeated interaction, or some idiosyncratic match-specific productivity, it will prevent all buyers from switching to the lowest-cost intermediate goods suppliers.

To provide some empirical evidence for the persistence of buyer-supplier relationships, I explore the shipments sent by cement and paperboard manufacturers in the 1993 and 1997 Commodity Flow Surveys. Again, from the Commodity Flow Survey, I cannot observe the identity of the downstream buyer. Instead, I proxy for the identity of the downstream buyer using the destination zip code. I run a conditional logit regression, described by Equation 37; the dependent variable equals 1 if the cement/paperboard plant, i, ships to zip code, z, in 1997. The explanatory variable of interest is an indicator, which equals 1 if the plant shipped to the zip code in 1993. Destination zip-code level fixed effects, supplier fixed effects, and the log distance between plant i and zip code z are additional explanatory variables.[49]

$$\begin{aligned}\mathbb{I}\{i \to z \text{ in } 1997\} = \beta_z + \beta_i + \beta_1 \log(\text{distance } i \to z) + \beta_2 \mathbb{I}\{i \to z \text{ in } 1993\} \\ + \beta_3 \mathbb{I}\{\text{plant of i's firm is located in } z \text{ in } 1997\} + \varepsilon_{iz}\end{aligned} \tag{37}$$

The results are presented in Table 13. Both cement and paperboard suppliers' decisions on which destinations to ship to are persistent across time. If plant i sells to zip code z in 1993, the probability that i will zip to z in 1997 is much higher, approximately 8 times larger for cement manufacturers and 10 times larger for paperboard manufacturers.

There are two distinct interpretations of the positive estimate on, β_2, the coefficient on the persistence of buyer-supplier relationships (see Heckman (1981), or Dubé, Hitsch, and Rossi (2010)).[51] In the first interpretation, an establishment's profitability of working with a counterparty increases from having transacted with that counterparty in the past. There are many reasons why the profitability of a buyer-supplier relationship might increase as buyers and suppliers interact with one another. Kellogg (2011), for instance, documents that oil

[49] Equation 37 can be derived from a discrete choice model in which establishments choose which zip code to sell to. Suppose that establishments can sell to at most one zip code and suppose that the profits that establishment i would earn from selling to zip code z are given by:

$$\begin{aligned}u_{izt} &= \alpha_i + \alpha_z - \alpha_1 \log(\text{distance } i \to z) \\ &\quad -\alpha_2 \mathbb{I}\{i \to z \text{ in year } t-1\} - \alpha_3 \mathbb{I}\{\text{plant of i's firm} \in z\} + \eta_{izt}.\end{aligned} \tag{36}$$

When η_{izt} is distributed according to the Type 1 Extreme Value distribution, establishments choices will be consistent with Equation 37.

[50] In addition, I restrict the sample to establishments that were sampled in both the 1993 and 1997 Commodity Flow Surveys. Secondly, in order to comply with Census disclosure rules, I restrict the sample to plants that are members of firms, f, such that the following three criteria hold: a) there exists at least one i, z pair for which plant i (owned by firm, f) shipped to zip code z in 1993, but not in 1997; b) there exists at least one i, z pair for which plant i shipped to zip code z in 1997, but not in 1993; and c) there exists at least one i, z pair for which plant i shipped to zip code z in both 1993 and 1997. The coefficient estimates are similar when the sample is constructed without this second restriction.

[51] Spurious state dependence occurs when the η_{izt} in Equation 36 are serially correlated, or if there are omitted variables in Equation 37.

Sample	Cement	Cement	Cement	Paperboard	Paperboard	Paperboard
Did the plant sell to		2.075	2.009		2.988	2.661
the zip code in 1993?		(0.105)	(0.105)		(0.067)	(0.069)
Log Mileage	-2.835	-2.594	-2.593	-1.025	-0.867	-0.881
	(0.050)	(0.050)	(0.051)	(0.024)	0.026	(0.026)
N	106795	106795	106795	75360	75360	75360
Number of zip codes	2015	2015	2015	1256	1256	1256
Number of plants	53	53	53	60	60	60
Pseudo- R^2	0.687	0.713	0.718	0.148	0.257	0.290
Unconditional probability of entry	0.021	0.021	0.021	0.030	0.030	0.030
Include control for firm presence in z?	No	No	Yes	No	No	Yes

Table 13: Coefficient estimates and standard errors, from the regression defined by Equation 37. The sample is comprised of cement and paperboard plants that were included in the sample of Regressions 34, 35, and 40. For a zip code to be in the sample, at least one plant must have shipped to the zip code.[50]

production companies and drillers become more productive as they gain experience working with one another. Alternatively, according to the second interpretation, some establishments happen to find it more profitable to work with certain counterparties for idiosyncractic reasons, other than geographic proximity. The estimate of the persistence term, β_2, will be positive provided these idiosyncratic factors display some persistence. Unfortunately, the data that I have at hand do not permit me to distinguish between these two interpretations. Either interpretation, however, is consistent with downstream establishments that decide to remain matched with high-marginal-cost suppliers.

5.4 Price discriminatory behavior of suppliers

A third explanation for price variation lies in differences in the relative bargaining power of the suppliers and buyers of any given material input. As the model in Section 2.2 shows, plants may have different relative bargaining strength because they occupy different positions within the buyer-supplier network. In this subsection, I document that suppliers charge different prices–for the same good–across destinations. I then show that these within-supplier price differences explain a large fraction of the dispersion in the prices that downstream plants pay for their intermediate inputs.

For each buyer-supplier relationship, I define the *within-supplier price deviation*, ψ_{hit},

as:

$$\psi_{hit} \equiv \log\left(\frac{P_{hit}^{CFS}}{P_{ht}^{out,CFS}}\right), \tag{38}$$

ψ_{hit} is the price that i pays for h's output, compared to the other plants that buy intermediate inputs from h; ψ_{hit} is positive provided plant i purchases its material inputs from h at a higher price than $P_{ht}^{out,CFS}$, the average output price of supplier, h.

Motivated by the right panel of Figure 1, Figure 2 decomposes the price distribution into two separate components. Any buyer-supplier-specific price, p_{hit}^{CFS}, is the sum of the supplier's average output prices, $p_{ht}^{out,CFS}$, and the within-supplier price deviation, ψ_{hit}. The price, p_{hit}^{CFS}, that a supplier charges a buyer for intermediate inputs can be, mechanically, high for one of two reasons: either the supplier has a high average price, $p_{ht}^{out,CFS}$, or the supplier charges i a higher price than its other customers (i.e., ψ_{hit} is large). For my sample of cement and paperboard manufacturers, the distributions of p_{hit}^{CFS}, ψ_{hit}, and $p_{ht}^{out,CFS}$ are depicted in Figure 2.[52] The standard deviation of p_{hit}^{CFS} is 0.35, larger than the standard deviation of suppliers' average output prices ($SD(p_{ht}^{out,CFS}) = 0.28$), and almost twice as large as the standard deviation of the within-supplier deviations ($SD(\psi_{hit}) = 0.19$).

The *price-discrimination coefficient,* ψ_{it}, measures the extent to which plant i pays its supplier a higher materials price than the other customers of its suppliers. It is a weighted average, over i's suppliers, of the ψ_{hit}:

$$\psi_{it} \equiv \frac{\sum_{h:h\to i}\omega_{hit}\psi_{hit}}{\sum_{h:h\to i}\omega_{hit}} = \frac{\sum_{h:h\to i}\omega_{hit}\left(p_{hit} - p_{ht}^{out,CFS}\right)}{\sum_{h:h\to i}\omega_{hit}} \tag{39}$$

I regress buyers' average materials prices on their suppliers' marginal costs and their own ψ_{it}.

$$\tilde{p}_{it}^{in,CFS} = \beta_{\text{division}} + \beta_1 \cdot \overline{tfpq}_{it} + \beta_2 \cdot \psi_{it} + \varepsilon_{it} \tag{40}$$

The regression results are given in Table 14. As in Table 9, the estimate of β_1 is positive and statistically significant: plants with low-marginal cost suppliers pay lower than average prices for their intermediate inputs. In addition, there is a strong, positive relationship between $\tilde{p}_{it}^{in,CFS}$ and ψ_{it}.

A positive, significant coefficient estimate of β_2 is not surprising. A mechanical relationship between ψ_{it} and $\tilde{p}_{it}^{in,CFS}$ exists, as higher-than-average-priced shipments will generate a large value for $\tilde{p}_{it}^{in,CFS}$ (see Equation 32) and a large value of ψ_{it} (see Equation 39). The more interesting calculations, which are given in Table 15, compute the fraction

[52] Figure 2 looks similar, whether one uses the sample of cement manufacturers, the sample of paperboard manufacturers, or the pooled sample of paperboard and cement manufacturers.

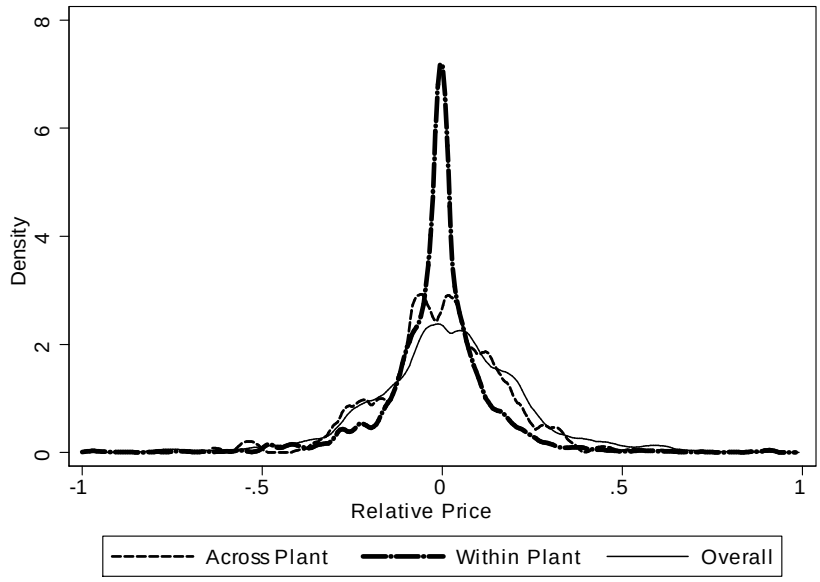


Figure 2: Value-weighted price distributions. The solid, thin line gives the distribution of $log(P_{hi}^{CFS})$. The thick, dash-dot line gives the disitribution of within-supplier price deviations, ψ_{hi}, while the thin, slashed line gives the distribution of suppliers' average output prices. The sample includes all shipments sent by cement and paperboard manufacturers that comprised the sample of the regressions defined by Equations 34 and 35.

Sample	Boxes	Concrete	Pooled	Boxes	Concrete	Pooled
$\overline{tfpq}_{it}$	-0.231*	-0.138	-0.210*	-0.229*	-0.144	-0.207*
	(0.060)	(0.085)	(0.051)	(0.056)	(0.103)	(0.047)
ψ_{it}	0.322*	0.868*	0.422*	0.326*	0.903*	0.419*
	(0.097)	(0.284)	(0.123)	(0.104)	(0.229)	(0.126)
N	190	131	321	190	131	321
Adjusted R^2	0.184	0.427	0.217	0.182	0.507	0.219
Division F.E.?	No	No	No	Yes	Yes	Yes

Table 14: Coefficient estimates and robust standard errors of the regressions defined by Equation 39. Stars indicate significance at the 5% level.

	Include Division Fixed Effects ?	No	No	Yes	No	Yes
	Include $\overline{tfq}_{it}$?	No	No	No	Yes	Yes
	Include ψ_{it} ?	No	Yes	Yes	Yes	Yes
Sample	Sample Size					
Boxes	190	0.202	0.191*	0.186	0.182*	0.178*
Concrete	131	0.374	0.286*	0.257*	0.281*	0.252*
Pooled	321	0.224	0.206*	0.202*	0.198*	0.195*

Table 15: Each cell gives the real-revenue-weighted standard deviation of the residuals in a particular regression; the full specification is given in Equation 40. Across the columns of the table, different combinations of independent variables are included in the regressions. Across the rows of the table, different samples are used. Stars indicate that the decline in dispersion is significantly more than the decline that would occur from simply including "fake" random variables on the right hand side of Equation 40. See Appendix E for details.

of the dispersion in $\tilde{p}_{it}^{in,CFS}$ that can be explained by ψ_{it}. Including ψ_{it} as a covariate reduces the unexplained variation of $\tilde{p}_{it}^{in,CFS}$ by 7% ($= \frac{0.224-0.206}{0.224}$). A combination of ψ_{it} and $\overline{tfpq}_{it}$ explains 12% ($= \frac{0.224-0.198}{0.224}$) of the variation of plants' materials prices, while the full combination of right-hand-side variables reduces the unexplained variation of $\tilde{p}_{it}^{in,CFS}$ by 13% ($= \frac{0.224-0.195}{0.224}$).

6 Conclusion

In this paper, I have studied the consequences and sources of materials price dispersion. Variation in materials prices explains a substantial fraction of the variation in plants' marginal costs, revenue total factor productivities, and probabilities of survival. Moreover, one reason why some plants have low materials prices is because they have access to suppliers with low marginal costs.

These results indicate that establishments' survival and growth prospects are directly related to those of their customers and/or suppliers. In future work, I hope to investigate the causal relationship between establishments' growth and the growth rates of their counterparties. Such an investigation will be an important building block in understanding the propensity with which shocks to a small set of firms have the potential to cascade throughout the economy and produce aggregate fluctuations.

Appendix

A Proofs

Proposition 1 (Reminded) When the revenue share, R_i, of each plant is sufficiently close to 0, the output price, P_i^{out}, is approximated by the following equation

$$\begin{aligned} \log P_i^{out} &\approx \frac{1}{2}\left(\log MC_i + \overline{\log MC}\right) + \frac{1}{2\gamma}\left(\Delta_i + \overline{\Delta}\right) \\ &\approx \frac{1}{2}\left(\sigma \log P_i^{in} - \log \Phi_i + \sigma\overline{\log P^{in}} - \overline{\log \Phi}\right) + \frac{1}{2\gamma}\left(\Delta_i + \overline{\Delta}\right) \end{aligned} \tag{41}$$

Proof.

Write the profits of firm i as the product of a) quantity b) and the difference between price and marginal costs:

$$\Pi_i = \frac{\Upsilon R_i}{P_i^{out}}\left(P_i^{out} - MC_i\right) = \Upsilon R_i\left(1 - \frac{MC_i}{P_i^{out}}\right)$$

Take first order conditions and re-arrange. Doing so results in a markup, $\frac{P_i^{out}}{MC_i}$, that equals $1 - \left[\frac{\partial \log R_i}{\partial \log P_i^{out}}\right]^{-1}$. Using the expression for $\frac{\partial \log R_i}{\partial \log P_i^{out}} = -\frac{I-1}{I}\frac{\gamma}{R_i}$ (which can computed from Equation 3), plant i's output price is given by the following equation:

$$\frac{P_i^{out}}{MC_i} = 1 + \frac{R_i I}{\gamma(I-1)} \tag{42}$$

Assuming that R_i is sufficiently close to 0, $\log\left(1 + \frac{R_i I}{\gamma(I-1)}\right)$ is approximately equal to $\frac{R_i}{\gamma}\frac{I}{I-1}$. Thus, Equation 42 leads to

$$\log P_i^{out} \approx \log MC_i + \frac{R_i}{\gamma}\frac{I}{I-1}. \tag{43}$$

Furthermore, the revenue share of plant i equals (see Equation 3)

$$R_i = \Delta_i - \frac{\gamma(I-1)}{I}\log P_i^{out} + \frac{\gamma}{I}\sum_{k=1, i\neq k}^{I} \log P_k^{out},$$

yielding:

$$\begin{aligned}\frac{R_i I}{\gamma(I-1)} &= \Delta_i \frac{I}{\gamma(I-1)} - \log P_i^{out} + \frac{1}{I-1}\sum_{k\neq i} \log P_k^{out} \\ &= \Delta_i \frac{I}{\gamma(I-1)} - \frac{I}{I-1}\log P_i^{out} + \overline{\log P^{out}}. \end{aligned} \tag{44}$$

Plugging Equation 43 into Equation 44 leads to

$$\begin{aligned}\log P_i^{out} &\approx \log MC_i + \Delta_i \frac{I}{\gamma(I-1)} - \frac{I}{I-1}\log P_i^{out} + \overline{\log P^{out}}. \\ \log P_i^{out} &\approx \frac{1}{2}\left(\log MC_i\right) + \frac{\Delta_i}{\gamma} + \frac{\overline{\log P^{out}}}{2} \end{aligned} \tag{45}$$

The equation

$$\frac{1}{I}\sum_{i=1}^{I} \log P_i = \frac{1}{2I}\sum_{i=1}^{I}\left(\log MC_i + \overline{\log P}\right) + \frac{1}{2\gamma}\frac{1}{I}\sum_{i=1}^{I}\Delta_i,$$

is equivalent to

$$\overline{\log P^{out}} \approx \overline{\log MC} + \frac{1}{\gamma}\overline{\Delta}. \tag{46}$$

Thus, plugging 46 into Equation 45 yields

$$\log P_i \approx \frac{1}{2}\left(\log MC_i + \overline{\log MC}\right) + \frac{1}{2\gamma}\left(\Delta_i + \bar{\Delta}\right),$$

which is the desired result. ■

Proposition 3 (Reminded) Assume that a) $\text{Cov}(\log P^{in}, \log \Phi)$,$\text{Cov}(\log P^{in}, \Delta)$, and $\text{Cov}(\Delta, \log \Phi)$ are sufficiently close to 0 in the observed sample, and b) $R_i \to 0$ for all plants i. Then, the following relationships hold:

1. The materials price is increasing, in the sense of first order stochastic dominance, in the marginal cost of the supplier.

2. Plants' marginal costs are positively correlated to their suppliers' marginal costs.

Proof.

1. Let $G(\cdot)$ denote the cumulative distribution function of suppliers' marginal costs, and let $H(P^{in}|MC^{up})$ denote the cumulative distribution function of the price that i pays

for the intermediate input, conditional on the marginal cost of the lowest-marginal-cost supplier.

Then:

$$\begin{aligned} 1 - H(P^{in}|MC^{up}) &= \Pr\left\{\xi - 1 \text{ draws each } \geq P^{in} | \xi - 1 \text{ draws each} \geq MC^{up}\right\} \\ &= \left(\frac{1 - G(P^{in})}{1 - G(MC^{up})}\right)^{\xi - 1} \end{aligned}$$

Re-arranging:

$$H(P^{in}|MC^{up}) = 1 - \left(\frac{1 - G(P^{in})}{1 - G(MC^{up})}\right)^{\xi - 1}$$

The desired result follows from the fact that the right hand side of the previous equation is decreasing in, MC^{up}, the marginal cost of the best supplier.

2. Since, from part 1, $\frac{\partial E\left[\log P^{in}|MC^{up}\right]}{\partial MC^{up}} > 0$, $Cor(\log P^{in}, \log MC^{up}) > 0$. Since I assumed that $Cov(\log MC^{up}, \log \Phi) \approx 0$

$$\begin{aligned} Cov(\log MC, \log MC^{up}) &= Cov(\sigma \log P^{in}, \log MC^{up}) - Cov(\log \Phi_i, \log MC^{up}) \\ &> 0 \end{aligned}$$

■

B Other Calculations

B.1 An expression for industry sales in terms of marginal costs and demand shocks

As a reminder, expenditures on the industry output equals (see Equation 2):

$$\log \Upsilon = \Delta_0 + \sum_{i=1}^{I} \Delta_i \log P_i^{out} - \frac{\gamma}{2} (\log P_i)^2 + \frac{\gamma}{2I} \sum_{i=1}^{I} \sum_{k=1}^{I} \log P_i^{out} \log P_k^{out}$$

This is equivalent to:

$$\begin{aligned} \log \Upsilon &= \Delta_0 + I \cdot E\left[\Delta \log P^{out}\right] - \frac{I\gamma}{2} E\left[\left(\log P^{out}\right)^2\right] + \frac{\gamma I}{2} \left(\overline{\log P^{out}}\right)^2 \qquad (47) \\ &= \Delta_0 + I \cdot E\left[\Delta \log P^{out}\right] - \frac{I\gamma}{2} Var\left(\log P^{out}\right) \end{aligned}$$

Next, I derive the following expression for $Var(\log P^{out})$, in terms of moments of the joint distribution of MC and Δ. Note that these moments are taken over the sample of plants that we observe selling positive amounts of the good.

$$\begin{aligned} Var(\log P^{out}) &= Var\left(\frac{1}{2}(\log MC) + \frac{1}{2\gamma}(\Delta)\right) \\ &= \frac{1}{4}Var(\log MC) + \frac{1}{4\gamma^2}Var(\Delta) + \frac{1}{2\gamma}Cov(\log MC, \Delta) \end{aligned} \tag{48}$$

Similarly, I derive the following expression for $E\left[\Delta \log P^{out}\right]$, in terms of the moments of the joint distribution of MC and Δ.

$$\begin{aligned} I \cdot E\left[\Delta \log P^{out}\right] &= \frac{I}{2}E\left[\Delta \log MC + \frac{\Delta^2}{\gamma}\right] + \frac{I\bar{\Delta}\overline{\log MC}}{2} + \frac{I\left(\bar{\Delta}\right)^2}{2\gamma} \\ &= \frac{I}{2}E\left[\Delta \log MC\right] + \frac{I}{2\gamma}E\left[\Delta^2\right] + \frac{\overline{\log MC}}{2} + \frac{\bar{\Delta}}{2\gamma} \end{aligned} \tag{49}$$

Finally, I plug Equations 48 and 49 into Equation 47, which leads to the following expression for Υ, in terms of the moments of the joint distribution of MC and Δ:

$$\begin{aligned} \log \Upsilon &= \Delta_0 + \frac{I}{2} \cdot Cov(\Delta, \log MC) + \frac{I}{2\gamma}Var(\Delta) + \frac{\overline{\log MC}}{2} \\ &\quad + \frac{\bar{\Delta}}{2\gamma} - \frac{I\gamma}{8}Var(\log MC) - \frac{I}{8\gamma}Var(\Delta) - \frac{I}{4}Cov(\log MC, \Delta) \\ &= \frac{I\gamma}{8}\left[2 \cdot Cov\left(\frac{\Delta}{\gamma}, \log MC\right) + 3 \cdot Var\left(\frac{\Delta}{\gamma}\right) - Var(\log MC)\right] \\ &\quad \Delta_0 + \frac{1}{2}\left[E\left[\log MC\right] + E\left[\frac{\Delta}{\gamma}\right]\right] \end{aligned}$$

The final expression for $\log \Upsilon$ is the value that appears in footnote 11.

C Construction of the Sample

The benchmark sample consists of 10 industries (collections of 7-digit products) for which both inputs and outputs display minimal levels of quality differentiation. The construction of the sample consists of plants for which the following five conditions hold. First, I discard any plants that have missing data on labor inputs, capital stocks, electricity bills, or materials bills. Second, I discard any plants that do not fill out either the Census of Manufacturers Materials Supplement (containing information on purchases of intermediate inputs) or the Census of Manufacturers Productivity Supplement (containing information

on products produced). Third, I throw out plants that have imputed values for quantities of materials purchased or products produced.[53] Fourth, I require that the plants in the benchmark sample earn at least half of their revenues from one of the 10 main industries. Fifth, I discard any plant that has an output price (defined by p^{out}, as in Equation 16), an input price (defined by p^{in}, as in Equation 17 or 19), or a quantity total factor productivity (defined by $tfpq$, as in Equation 20) that is more than 3 units away than the average for that industry×year.

Industries are defined as the collection of 7-digit products in the following manner.

Bulk milk is the combination of fluid whole milk, bulk sales (2026112) and fluid skim milk, bulk sales (2026115). The units of bulk milk are thousands of pounds.

White wheat flour is the combination of the ten 7-digit products: white flour, shipped for export (2041105 and 2041107); bakers' and institutional white bread-type flours (2041111 and 2041113); bakers' and institutional soft wheat flour (2041115 and 2041117); family white flour, other than self-rising (2041121 and 2041123); self-rising family white flour (2044126); and flour shipped to blenders or other processors (2041128 and 2041129). The units of white wheat flour are 50-pound sacks.

Ready-mix concrete consists of the single 7-digit product (3273000). In 1972 and 1977 some ready-concrete plants were producing a product with a code of 3273011. The units of output are cubic yards. Production data do not exist for 1997; materials data do not exist for 1992 or 1997. Because of this, for the analysis in Section 4, the sample period for ready mix concrete is 1972-1987. The sample period for the analysis of Section 5, in which I use the Commodity Flow Survey but not the Census of Manufacturers' materials data, is 1992. In addition to the five criteria listed in the first paragraph of this section, I require ready-mix concrete plants to have positive purchases of both cement and sand/gravel. (I need positive purchases of both of the main intermediate inputs so that I can estimate the elasticity of substitution between cement and sand/gravel, using Equation 50.)

Yarn is comprised of the two 7-digit products, spun gray (22811100) and yarn, spun

[53] White, Reiter, and Petrin (2012) argue that, because of survey nonresponse, on average, 40% of the non-adminisrative record plants in the Census of Manufacturers have imputed data. Moreover, because the Census uses industry averages to impute missing values for shipments, materials purchases, or other variables, the imputation method causes a downward bias in estimated within-industry productivity dispersions, and also biases the measured relationships among plant-level characteristics. With this in mind, I have chosen to exclude all plants with imputed data on the quantities of materials purchases or goods shipped. (Unfortunately, imputed-data flags for other variables–employment, electricity purchases, etc...–exist only beginning in 2002. However, using data from 2002, I have checked that there are very few observations with a) non-imputed quantities of materials/output and b) imputed values for other relevant variables. For 2002, I have also checked that the difference–between the three productivity measures– is robust to the inclusion/exclusion of observations that have imputed values for the "other" variables.) Then, at least for my selected sample, I will be able to accurately measure within-industry dispersions of prices and productivities.

and finished in the same establishment (2281187). The units of output are thousands of pounds.

Gasoline is comprised of the following three 7-digit products: motor gasoline (2911131), distillate fuel oil (2911412), and No. 4 type light fuel oil (2911414). The units of output are thousands of barrels.

Coffee consists of two 7-digit products, whole bean coffee (2095111) and ground coffee (2095115). The units of output are thousands of pounds.

Sugar consists of the single 7-digit product, raw cane sugar (2061011). The units of output are short tons.

Packaged milk consists of the following three 7-digit products: fluid whole milk (2026212), low fat milk (202623), and skim milk (2026225). The units of output are thousands of quarts.

Corrugated boxes is a combination of nine 7-digit products, with products being classified by their end use. These end-uses are containers of food and beverages (2653012); carry-out boxes for retail food (2653014); containers of paper and allied products (2653013); containers of glass, clay, and stone products (2653015); containers of metal products, machinery, equipment, and supplies (2653016); containers of electrical machinery, equipment, supplies, and appliances (2653018); containers of chemicals and drugs, including paints, varnishes, cosmetics, and soaps (2653021); containers of lumber and wood products, including furniture (2653022); all other end uses not specified (2653030). From 1972 to 1987, the units of output for corrugated boxes were thousands of pounds. From 1992 on, the units of output for corrugated boxes have been thousands of square feet.

Measuring corrugated boxes in terms of area, instead of mass, is somewhat problematic. The density of boxes depend on their final use. In particular, the density of boxes is lower for those boxes that are used as containers of food, beverages, paper and allied products, glass, clay, stone, or metal, while the density of boxes is higher for boxes that are used as containers of machinery, electronics, chemicals, lumber, and other products. Since the total cost of producing corrugated boxes seems to be more closely related to the mass–instead of surface area–of the amount produced, measured quantity total factor productivity for low-density box manufacturers began to exceed, in 1992, the measured quantity total factor productivity of high-density boxes.

To mitigate the impact of this measurement problem, I de-meaned, according to Equation 23, plant-level statistics separately for the high-density (those plants that produced output with a product code between 2653016 and 2653030) and low-density (those plants that produced output with a product code between 2653012 and 2653015) box manufacturers.[54]

[54]Dropping the "Boxes, Year≥1992" subsample does not change any of the results from Section 4. I find it

	Employment		Total Value of Shipments		N		Main
Sample	Benchmark	Main Ind.	Benchmark	Main Ind.	Benchmark	Main Ind.	Ind.
Bulk Milk	3.950	3.441	9.082	8.023	127	7661	2026
Flour	4.763	3.010	9.854	7.942	503	2073	2041
Concrete	3.547	2.350	7.682	6.827	3708	20956	3273
Yarn	5.942	4.749	9.508	8.645	431	2233	2281
Boxes, Yr.$\leq$'87	5.399	3.998	9.426	8.404	1820	7742	2653
Gasoline	6.554	4.867	12.977	11.157	692	1706	2911
Coffee	4.906	3.484	9.920	8.736	300	874	2095
Sugar	5.708	4.901	10.000	9.301	177	301	2061
Boxes, Yr.$\geq$'92	5.559	3.998	9.799	8.404	646	7742	2653
Packaged Milk	5.119	3.441	9.465	8.023	2099	7661	2026
Pooled	4.740	3.128	9.119	7.689	10503	43546	

Table 16: Descriptive statistics for the benchmark sample. Variables are stated in logs. The final column refers to the 4-digit SIC industry of which the product is a memeber.

In Table 16, I provide some descriptive statistics of the benchmark sample. The average log employment for plants is 3.93 (i.e., roughly 51 $\approx e^{3.93}$ employees work in the average plant.) Plants that produce ready-mix concrete are one-third the size of the average benchmark-sample plant, while plants engaged in gasoline production employ approximately 6 ($\approx e^{5.741-3.934}$) times as many workers as the average plant.

Compared to the universe of plants that are in the same 4-digit SIC industry, the plants in the benchmark sample employ 5.0($\approx e^{4.740-3.128}$) times as many employees and have revenues that are 4.2($\approx e^{9.12-7.68}$) times larger. The difference is due to the Census Bureau's survey methodology: the largest plants tend to receive the survey questionnaires on the products they produce or the materials they consume.

For a particular intermediate input to be included in the analysis, expenditures of the material must make up at least 6% of total materials expenditures for that product group. As the cut-off expenditure share decreases, additional intermediate inputs are included in the analysis. Setting the cut-off too low results in the inclusion of intermediate inputs that are purchased only by a few plants, hindering cross-plant comparisons of materials prices. Setting the cut-off too high means that important components of plants' materials prices are ignored. The 6% cut-off seems like a good compromise between these two considerations.

In some instances for which I combine groups of similar 6-digit products to form a given "material input".[55] For example, I combine material 131111 (domestic crude petro-

worth the trouble to keep the "Boxes, Year$\geq$1992" subsample, since corrugated box manufacturers purchase one of their main inputs–namely, paperboard–from the manufacturing sector, and thus can be included in the analysis of Section 5.

[55] For 1992 and 1997, a description of the 6-digit material codes can be found by downloading MC92F7.dbf

leum) and 131112 (foreign crude petroleum). The presumption when deciding to combine two materials is that the manufacturer is indifferent between the two 6-digit materials. The way in which I combined these 6-digit products is given below.

For milk (either bulk or packaged), the sole material is unpasteurized whole milk (024111).

For white wheat flour, the sole material is wheat (011111).

In the production of ready-mix concrete, the two materials are cement (which was coded as 324101 in 1982 and 1992 and 324102 in other years) and sand/gravel aggregate (144201).

In the production of yarn, the two materials are raw cotton fibers (013101) and a combination of polyester staple and tow (282425) and acrylic staple and tow (282426).

In the production of gasoline, I have combined foreign and domestic crude petroleum into one material.

In the production of raw cane sugar, the sole material is sugar cane (013321).

Green coffee beans (017921) are the sole material used in the production of ground/whole bean coffee.

Finally, in 1992-1997, the sole material used in the production of corrugated boxes is coded 260003 ("Paper and Paperboard"). In 1987, the material "Paper/Paperboard" is the combination of 262104 ("Paper, Cellulosic Wadding") and 262108 (Paper). Earlier than this, "Paper/Paperboard" is the combination of materials 262102, 262103, and 262105.

D Robustness Checks

D.1 Industries with heterogeneous quality outputs

In this subsection, I reproduce the empirical analysis of Sections 4.1 and 4.2 for a set of industries that display substantial output quality variation. The four industries that I chose for this exercise are wine, softwood cut stock, cucumber pickles, and sausages.

Industries are defined as the collection of 7-digit products in the following manner:

Wine is a combination of the following three products: white grape wine (2084012), red grape wine (2084014), and rose grape wine (2084016). The units of output are thousands of gallons.

Softwood cut stock is a combination of two product groups: furniture cut stock (2421711) and industrial cut stock (2421751). The units of output are thousands of board feet.

from the following Census web-page: ftp://ftp2.census.gov/econ1992/MC92/.

Sample	Units of Output	Material Inputs	N
Wine	1000 gallons	Fresh grapes (41%) Purchased wines (23%) Glass containers (19%)	330
Softwood Cut Stock	1000 board feet	Softwood dressed lumber (75%) Softwood logs (8%) Hardwood dressed lumber (8%)	160
Pickles	1000 gallons	Cucumbers (43%) Glass containers (28%)	145
Sausages	1000 lbds.	Fresh and frozen pork (34%) Fresh and frozen beef (30%) Meat, unknown species (13%)	621
Pooled	-		1256

Table 17: Description of the four industries comprising the "Quality Variation Sample". The Material Inputs column gives the inputs that represent greater than 6% of the average plants' total material purchases. The percentages that appear in the Material Inputs column are the fraction of intermediate input expenditures that go to each particular material input.

Pickles are a combination of four products: dill pickles (2035211), sour pickles (2035213), sweet pickles (2035215), and refrigerated pickles (2035219). The units of output are thousands of gallons.

Sausages are a combination of six products: fresh sausage (2011711 and 2013711); dry or semi-dry sausages (2011717 and 2013717); and frankfurters (2011721 and 2013721). The units of output are thousands of pounds.

As with the benchmark sample, materials that make up more than 6% of intermediate input expenditures are included as "priced" materials. A summary of the characteristics of the Output Quality Variation sample are given in Table 17.

Correlations among plant-level characteristics are presented in Table 18. Compared to the benchmark sample, the standard deviations of most plant-level characteristics are larger, while the skewness coefficients are smaller. (A positive skewness coefficient of, for example, $tfpr$ could reflect the existence of a productivity threshold $\widetilde{tfpr}$, below which no establishments may profitably operate.) In other words, when outputs vary in quality, the distribution of plant-level characteristics both becomes less skewed and more disperse. Moreover, compared to the benchmark sample, the correlations among the different productivity measures are weaker.

The dispersions of $tfpq$ and ϕ are given in Table 19. For the pooled sample, the dispersions of the two distributions are essentially the same. Looking across the four industries, there is a significant decline in productivity dispersion for one of the industries,

	p^{in}	p^{out}	$tfpq$	ϕ	$tfpr$
p^{out}	0.284*				
$tfpq$	-0.273*	-0.808*			
ϕ	0.254*	-0.659*	0.858*		
$tfpr$	0.024	0.329*	0.305*	0.290*	
Std. Dev.	0.318	0.385	0.380	0.380	0.237
Skewness	0.276	0.051	0.070	0.273	0.777

Table 18: Correlations among plant-level characteristics, pooled across the four industries of the "Quality Variation Sample." N=1256. Stars indicate that the correlation is significantly different from 0, at the 5% level (see Appendix E for details).

	Dispersion of $tfpq$			Dispersion of ϕ			Percent Decline			
Sample	90/10	75/25	SD	90/10	75/25	SD	90/10	75/25	SD	N
Wine	1.314	0.696	0.479	1.268	0.726	0.481	3.6%	-4.0%	-0.5%	330
Softwood Cut S.	1.571	0.818	0.547	1.316	0.697	0.477	21.4%*	18.9%	16.0%*	160
Pickles	0.891	0.446	0.344	0.881	0.412	0.339	1.2%	8.6%	1.3%	145
Sausages	0.727	0.399	0.295	0.747	0.366	0.301	-2.7%	9.5%	-2.0%	621
Pooled	0.948	0.494	0.380	0.977	0.486	0.380	-2.9%	1.7%	0.0%	1256

Table 19: Dispersion of $tfpq$ and ϕ, for the "Quality Variation Sample." Observations are weighted by revenue. Stars indicate that he difference between $tfpq$ and ϕ is statistically significant, at the 5% level (see Appendix E for details). Due to Census' rules regarding data confidentiality, I am prohibited from reporting the actual quantiles of any empirical distribution. See the caption of Table 3 for a description of the imputation of the quantiles of $tfpq$ and ϕ.

softwood cut stock, and no difference for the other three industries.

In summation, output quality variation has the potential to severely attenuate the difference between the dispersions of ϕ and $tfpq$. To the extent that any quality variation exists in the benchmark sample, the difference between the dispersions of ϕ and $tfpq$, as reported in Table 3, may be downwardly biased.

D.2 Variation in input quality

One of the main presumptions of the empirical analysis is that variation in input quality is not an important source of variation of input prices or productivities. I have chosen industries to try to minimize the role of input quality differentiation. There is one specific industry, ready-mix concrete, for which there is reason to suspect that input quality differences could be contaminating some of the results. In this subsection, I explain the reason why input quality varies across plants, and then determine how big of an effect input quality variation has on the observed relationships between input prices and different productivity

measures.

Portland cement, the main intermediate input used in the production of ready-mix concrete, comes in four types, labeled type I, II, III, or IV.[56] Sales of type-I and II cement constitute over 90% of the expenditures on cement, with the majority of sales coming from type-I cement (U.S. Department of Interior (1989)). In areas where the soil has high sulfate concentrations, type-II cement may be preferable to type-I cement, since ready-mix concrete produced using type-I cement is susceptible to *sulfate attack* (cracking or loss of strength in the presence of sulfate). Since high-sulfate concentrations exist only in the soil of parts of the western third of the United States, one should observe type-I and type-II cement consumed in the western United States, and only type-I cement consumed in the remainder of the United States.[57]

So, because of differences in soil composition, input quality variation should be greater in the western United States than in the eastern United States. With this in mind, I split the sample of ready-mix concrete plants into two subsamples: plants residing in Census divisions 1-7, and plants located in Census divisions 8-9.[58] In the left panel of Figure 3, I plot the distribution of p^{in} separately for the two halves of the United States. As expected, ready-mix concrete plants in divisions 8 and 9 pay a higher price for their intermediate inputs, as some of these plants pay a higher price to purchase type-II Portland cement. Moreover, the dispersion of p^{in} is larger in divisions 8-9, as some ready-mix concrete plants purchase the low-price type I cement, while others must purchase the high-price type-II cement. In contrast, in the eastern United States, virtually all ready-mix concrete plants purchase type-I cement, leading to a more compressed p^{in} distribution.

In the right panel of Figure 3, I plot the relationship between p^{in} and $tfpq$, separately for the two subsamples. The negative relationship between $tfpq$ and p^{in} is somewhat stronger in the eastern United States, while the correlations between p^{in} and p^{out} is greater in the western United States. These geographic differences are consistent with greater cement quality variation in the western United States.

Finally, in Table 21, I compute the dispersion of $tfpq$ and ϕ for the ready-mix concrete subsamples. The decline in dispersion is larger for each of the two subsamples than it is for the pooled sample of 3708 ready-mix concrete plants.

[56] The standards for the different types of Portland cement are set by the American Society for Testing and Materials (ASTM). See the ASTM webpage for more information on the distinguishing factors of different types of Portland cement: http://www.astm.org/Standards/C150.htm

[57] I confirm this prediction using the Census of Manufacturers production file.

[58] Census division 8 is made up of Arizona, Colorado, Idaho, Montana, Nevada, New Mexico, Utah, and Wyoming, while Census division 9 is made up of Alaska, California, Hawaii, Oregon, and Washington. See http://www.census.gov/geo/www/us_regdiv.pdf for a correspondence between states and Census divisions.

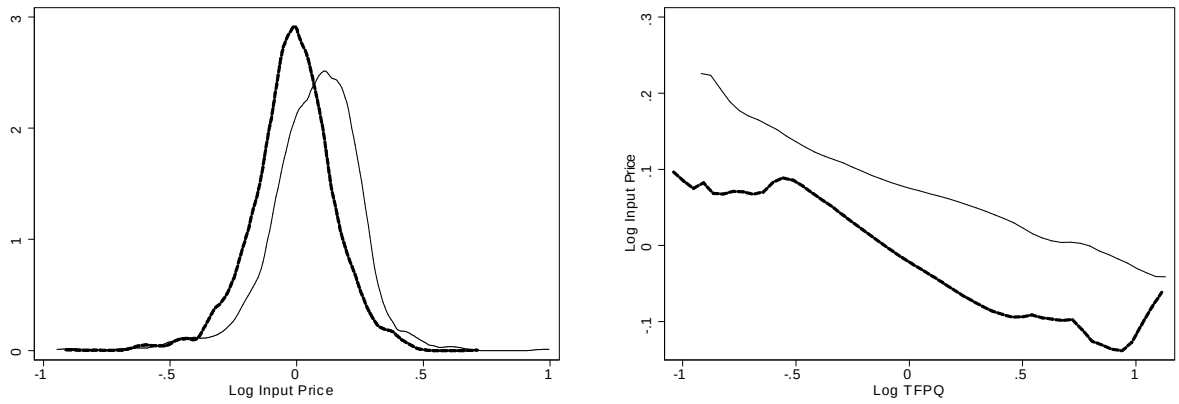


Figure 3: In the left panel, I plot the distribution of p^{in} separately for ready-mix concrete platns located in divisions 1-7 (thick, dashed line) and plants located in divisions 8-9 (thin, solid line). In the right panel, I plot the relationship between $tfpq$ and p^{in}, again separately for divisions 1-7 and 8-9.

Sample	$\rho(p^{in}, tfpq)$	$\rho(p^{in}, \phi)$	$\rho(p^{in}, p^{out})$	$\rho(\phi, tfpq)$	$\rho(tfpq, tfpr)$	$\rho(tfpq, p^{out})$	N
Entire U.S.	-0.306*	0.120*	0.276*	0.908*	0.741*	-0.476*	3708
Div. 1-7	-0.341*	0.089*	0.251*	0.906*	0.750*	-0.430*	3049
Div. 8-9	-0.351*	0.066	0.360*	0.911*	0.688*	-0.597*	659

Table 20: Correlations between different plant-level statistics of ready-mix concrete producers. Each row gives the correlations for different subsamples of ready-mix concrete plants.

	Dispersion of $tfpq$			Dispersion of ϕ			Percent Decline			
	90/10	75/25	SD	90/10	75/25	SD	90/10	75/25	SD	N
Entire U.S.	0.521	0.251	0.224	0.486	0.236	0.215	7.4%*	6.6%*	4.4%*	3708
Divisions 1-7	0.486	0.238	0.211	0.439	0.222	0.199	11.3%*	7.7%*	6.2%*	3049
Divisions 8-9	0.611	0.296	0.255	0.570	0.253	0.239	7.4%	18.3%*	6.9%*	659

Table 21: Dispersion of $tfpq$ and ϕ, for different subsamples of plants in the ready-mix concrete industry. Stars indicate that he difference between $tfpq$ and ϕ is statistically significant, at the 5% level (see Appendix E for details). Due to Census' rules regarding data confidentiality, I am prohibited from reporting the actual quantiles of any empirical distribution (e.g. ϕ). See the caption of Table 3 for a description of the imputation of the quantiles of $tfpq$ and ϕ.

In summation, there is almost no variation in the quality of cement purchased by ready-mix concrete plants in the eastern two-thirds of the United States. For this subsample, the difference between the standard deviation of $tfpq$ and the standard deviation of ϕ is 2 to 3 percentage points larger than the differences that are reported in Table 3. So, moderate amounts of intermediate input quality variation is probably causing me to somewhat underreport the fraction of productivity dispersion that is due to differences in materials prices.

D.3 Substitution across material inputs

Throughout the body of the paper, I assume that the elasticity of substitution between different material inputs–for industries that use multiple material inputs–is 0 (see Assumption 4 on page 19). For plants that produce ready-mix concrete, I assess the importance of the assumption that plants may not substitute across different material inputs.

There is some evidence that ready-mix concrete manufacturers substitute across the two intermediate inputs. I regress plants' utilization of cement, relative to sand/gravel, against the ratio of plant-level cement and sand/gravel prices. The results, which are given in Table 22, indicate that the elasticity of substitution is significantly greater than 0.[59]

$$\log\left(\frac{N_{ijt}^{Cement}}{N_{ijt}^{Sand/Gravel}}\right) = \beta_{\text{division}\times\text{year}} + \beta \cdot \log\left(\frac{P_{ijt}^{Cement}}{P_{ijt}^{Sand/Gravel}}\right) + \varepsilon_{ijt} \tag{50}$$

When the elasticity of substitution between gravel/sand and cement is constant (but not necessarily 0), the price of a bundle of material inputs equals:

$$P_{ijt}^{in} \equiv \left[\frac{s_{Gravel,jt}}{s_{Gravel,jt}+s_{Cement,jt}}\left(\frac{P_{Gravel,ijt}^{in}}{\bar{P}_{Gravel,jt}^{in}}\right)^{1-\varrho} + \frac{s_{Cement,jt}}{s_{Gravel,jt}+s_{Cement,jt}}\left(\frac{P_{Cement,ijt}^{in}}{\bar{P}_{Cement,jt}^{in}}\right)^{1-\varrho}\right]^{\frac{1}{1-\varrho}} \tag{51}$$

In Equation 51, $s_{Gravel,jt}$ refers to the share of materials expenditures that go to gravel, $P_{Gravel,ijt}^{in}$ is the price that plant i pays per 1000 pounds of gravel in year t, $\bar{P}_{Gravel,ijt}^{in}$ is the

[59] A common problem when running regressions similar to Equation 50 is the presence of simultaneous-equation bias: If a particular concrete manufacturer has a production technology that uses cement exceptionally intensively, the cement supplier may respond by increasing the price that it charges. The simultaneous-equation bias would case my estimated of β to under-represent the actual elasticity of substitution.

In unreported regressions, I instrument for $\frac{P_{ijt}^{Cement}}{P_{ijt}^{Sand/Gravel}}$ using local employment in the limestone industry (limestone is an intermediate input in cement manufacturing, but not in sand/gravel). The first-stage regressions are consistent with a negative relationship between local limestone employment and $\frac{P_{ijt}^{Cement}}{P_{ijt}^{Sand/Gravel}}$. The estimate of β, while substantially larger, is much less precisely estimated.

Fixed Effects	Sample	β	$s.e.$	Adjusted R^2	N
Year	Entire U.S.	-0.178	0.019	0.026	3708
Year×Division	Entire U.S.	-0.158	0.021	0.135	3708
Year×County	Entire U.S.	-0.209	0.035	0.467	3708
Year	Divisions 1-7	-0.092	0.022	0.010	3049
Year×Division	Divisions 1-7	-0.131	0.024	0.068	3049
Year×County	Divisions 1-7	-0.184	0.043	0.436	3049

Table 22: Each row presents the results of a regression, defined by Equation 50. The first three rows give estimates of the elasticity of substitution, using data from the entire U.S., while the last three rows give estimates computed using data only from the geographic divisions in which only type-I cement is used.

Sample	Revenue-weigthed?	90/10	SD	90/10	SD	90/10	SD	N
ϱ		0.1	0.1	0.3	0.3	0.5	0.5	
Concrete	No	0.5223	0.2302	0.5223	0.2302	0.5222	0.2303	3708
	Yes	0.4864	0.2151	0.4863	0.2151	0.4862	0.2151	3708
Pooled	No	0.4930	0.2190	0.4929	0.2190	0.4928	0.2191	10503
	Yes	0.3240	0.1506	0.3240	0.1506	0.3240	0.1506	10503

Table 23: Dispersion of ϕ, which is computed using Equations 22 and 23. Due to Census' rules regarding data confidentiality, I am prohibited from reporting the actual quantiles of any empirical distribution (e.g., ϕ). See the caption of Table 3 for a description of the imputation of the quantiles of $tfpq$ and ϕ.

geometric average of the price paid by all ready-mix concrete producing plants in year t, and ϱ is the elasticity of substitution between cement and sand/gravel. In the baseline analysis, I had set $\varrho = 0$.

Using Equation 51, I compute ready-mix concrete plants' materials prices. I then re-compute Φ_{ijt}, using Equation 22, and ϕ_{ijt}, using Equation 23. The dispersion of ϕ is given in Table 23. As ϱ increases, the price of a bundle of intermediate inputs decreases for plants that have exceptionally cheap input prices for one of the two intermediate inputs. Also, as ϱ increases, the relative price of the bundle increases for plants that pay roughly the same relative price for the two intermediate inputs. It turns out that, in combination, these two effects have almost no impact on the overall dispersion of ϕ.

D.4 Substitution between material inputs and other inputs

The empirical analysis of Section 4 invokes the assumption that the elasticity of substitution between material inputs and all other inputs, which I will denote $\hat{\varrho}$, equals 1 (see Assumption 1 on page 17). In reality, material inputs are likely to be complements to

Revenue-weigthed?	0.2	0.4	0.6	0.8	1.0
No	0.1507	0.1507	0.1507	0.1507	0.1506
Yes	0.2193	0.2192	0.2192	0.2191	0.2191

Table 24: Dispersion of ϕ, which is computed using Equations 53 and 23. The dispersion of ϕ, when $\hat{\varrho} = 1.0$, equals the value given in the final row of Table 3. N=10,503.

other inputs. In this subsection, I analyze how the dispersion of ϕ differs under different assumptions on $\hat{\varrho}$.

Consider a plant with technical efficiency Φ_{ijt}. Assume that, for plant i, the price of a unit of the "priced" intermediate input is P^{in}_{ijt}. The price of the other inputs, P^{other}_{jt}, are assumed to be the same for all plants in the industry-year (see Assumption 2 on page 17). With an elasticity of substitution of $\hat{\varrho}$, plant i's marginal cost equals:

$$\begin{aligned} MC_{ijt} &= \frac{1}{\Phi_{ijt}} \left[S_{jt} \cdot \sigma_{jt} \cdot \left(\frac{P^{in}_{ijt}}{\bar{P}^{in}_{ijt}} \right)^{1-\hat{\varrho}} + (1 - S_{jt} \cdot \sigma_{jt}) \left(\frac{P^{other}_{jt}}{P^{other}_{jt}} \right)^{1-\hat{\varrho}} \right]^{\frac{1}{1-\hat{\varrho}}} \\ &= \frac{1}{\Phi_{ijt}} \left[S_{jt} \cdot \sigma_{jt} \cdot \left(\frac{P^{in}_{ijt}}{\bar{P}^{in}_{ijt}} \right)^{1-\hat{\varrho}} + 1 - S_{jt} \cdot \sigma_{jt} \right]^{\frac{1}{1-\hat{\varrho}}} \end{aligned} \tag{52}$$

As in Section 4, $\sigma_{jt} \cdot S_{jt}$ refers to the expenditure share of "priced" materials. Equation 52 states that plants' marginal costs are determined by their technical efficiencies (Φ_{ijt}) and the composite price that they face for intermediate inputs and other inputs. The elasticity, $\hat{\varrho}$, dictates how the prices of intermediate inputs and other inputs are combined. As $\hat{\varrho}$ decreases, a larger weight is allotted to the input with a higher relative price.

Re-arranging Equation 52 yields the following expression for Φ_{ijt} in terms of $TFPQ_{ijt}$ and P^{in}_{ijt}:

$$\Phi_{ijt} = TFPQ_{ijt} \left[S_{jt} \cdot \sigma_{jt} \cdot \left(\frac{P^{in}_{ijt}}{\bar{P}^{in}_{ijt}} \right)^{1-\hat{\varrho}} + 1 - S_{jt} \cdot \sigma_{jt} \right]^{\frac{1}{1-\hat{\varrho}}} \tag{53}$$

For the pooled benchmark sample, I use Equation 53 to compute the standard deviation of ϕ_{ijt}, for $\hat{\varrho} \in \{0.2, 0.4, .0.6, 0.8, 1.0\}$. These results, which are presented in Table 24, illustrate that the dispersion of ϕ is robust to changes in the elasticity of substitution, even as $\hat{\varrho}$ approaches 0. Varying the elasticity of substitution, $\hat{\varrho}$, only has a noticeable effect on the measured technical efficiency for plants that have very small or very large values of $P^{in}_{ijt} \div \bar{P}^{in}_{ijt}$. Since most plants have materials prices that are close to the industry average, $\hat{\varrho}$ does not substantially alter the measured dispersion of ϕ.

Sample	$\rho(p^{in},tfpq)$	$\rho(p^{in},\phi)$	$\rho(p^{in},p^{out})$	$\rho(\ \phi,tfpq)$	$\rho(\ tfpq,tfpr)$	$\rho(tfpq,p^{out})$
Milk-Bulk	-0.424*	-0.149	0.410*	0.958*	0.444*	-0.770*
Flour	-0.394*	0.468*	0.312*	0.628*	0.128	-0.722*
Concrete	-0.306*	0.120*	0.276*	0.908*	0.740*	-0.476*
Yarn	-0.301*	0.197	0.211*	0.875*	0.491*	-0.763*
Boxes, Yr.$\leq$'87	-0.286*	0.226*	0.280*	0.868*	0.418*	-0.797*
Gasoline	-0.395*	0.141*	0.171*	0.854*	0.824*	-0.368*
Coffee	-0.485*	0.030	0.343*	0.855*	0.592*	-0.584*
Sugar	-0.034	0.481*	0.237*	0.860*	0.858*	-0.466*
Boxes, Yr.$\geq$'92	-0.352*	0.089	0.286*	0.901*	0.116*	-0.877*
Milk-Packaged	-0.281*	0.049	0.225*	0.945*	0.435*	-0.754*
Pooled	-0.369*	0.127*	0.231*	0.873*	0.694*	-0.551*

Table 25: Correlations between plant-level characteristics. All variables are de-meaned by year and product. Stars indicate that the correlation is significantly different from 0, at the 5% level (see Appendix E for details).

D.5 More correlations

Table 25, below, presents correlations between plant-level characteristics for each of the 10 industries in the benchmark sample.

For several of the correlations, the subsample of raw-cane sugar manufacturing plants is anomalous. For this industry, plants' marginal costs are unrelated to their materials prices. Moreover, the correlation between input prices and technical efficiencies is much stronger (48%) than for most other subsamples. These patterns are somewhat puzzling. Most likely, the goods produced by raw cane sugar manufacturers is not as homogeneous as Foster, Haltiwanger, and Syverson (2008) and I have presumed.

Except for the raw-cane sugar industry, correlations among plant-level characteristics are qualitatively similar across the different industries in the benchmark sample. The correlation between materials prices and marginal costs is moderately negative for the 9 other industries, while the correlation between marginal costs and output prices is strongly negative (ranging between -37% and -88%). Finally, the three productivity measures are always highly correlated with one another, with the correlation between ϕ and $tfpq$ being larger than the correlation between $tfpq$ and $tfpr$.

D.6 Unweighted results

In this section, I present the unweighted versions of Tables 2, 3, 19, and 25. In the benchmark calculations, observations are revenue-weighted. To preview the main results, all of the main conclusions of Section 4 are robust to the weighting scheme.

	p^{in}	p^{out}	$tfpq$	ϕ	$tfpr$
p^{out}	0.278*				
$tfpq$	-0.303*	-0.653*			
ϕ	0.141*	-0.549*	0.899*		
$tfpr$	-0.098*	0.212*	0.601*	0.581*	
Std. Dev.	0.167	0.186	0.227	0.219	0.176
Skewness	0.497	0.169	0.279	0.446	0.848

Table 26: Correlations and sample statistics of plant-level characteristics, pooled across all products. All variables are de-meaned by year and product. N=10,503. Correlations give equal weight to all plant-year observations. Stars indicate that the correlation is significantly different from 0, at the 5% level (see Appendix E for details). Also, see Table 2 for the real-revenue-weighted version of this table.

Sample	$\rho(p^{in},tfpq)$	$\rho(p^{in},\phi)$	$\rho(p^{in},p^{out})$	$\rho(\phi,tfpq)$	$\rho(tfpq,tfpr)$	$\rho(tfpq,p^{out})$
Milk-Bulk	-0.280*	0.033	0.382*	0.950*	0.522*	-0.789*
Flour	-0.344*	0.454*	0.256*	0.681*	0.254*	-0.715*
Concrete	-0.271*	0.107*	0.234*	0.928*	0.776*	-0.458*
Yarn	-0.354*	0.117	0.297*	0.887*	0.455*	-0.788*
Boxes, Yr.$\leq$'87	-0.406*	0.153*	0.404*	0.841*	0.425*	-0.824*
Gasoline	-0.353*	0.203*	0.125*	0.844*	0.840*	-0.396*
Coffee	-0.293*	0.227*	0.300*	0.862*	0.517*	-0.645*
Sugar	0.055	0.459*	0.116	0.912*	0.889*	-0.405*
Boxes, Yr.$\geq$'92	-0.428*	0.142*	0.366*	0.834*	0.127*	-0.873*
Milk-Packaged	-0.282*	0.054*	0.237*	0.943*	0.456*	-0.753*
Pooled-Bench.	-0.303*	0.141*	0.278*	0.899*	0.601*	-0.653*

Table 27: Correlations between plant-level characteristics. All variables are de-meaned by year and product. Correlations give equal weight to all plant-year observations. Stars indicate that the correlation is significantly different from 0, at the 5% level (see Appendix E for details). See Table 25 for the real-revenue-weighted version of this table.

The first two tables, Tables 26 and 27, give the correlations among plant-level statistics. Overall, the correlation between p^{in} and p^{out} is somewhat larger, while the correlation between p^{in} and $tfpq$ is somewhat closer to 0.

Compared to the revenue-weighted versions, the unweighted dispersions of $tfpr$, $tfpq$, and ϕ are larger (see Table 28 for the benchmark sample and Table 29 for the "Output Quality Variation" sample). The larger dispersions have two sources. First, weighting by revenue gives more importance to high revenue-per-plant industries. Since gasoline, which by far has the largest average sales per plant, has more compressed $tfpr$, $tfpq$ and ϕ distributions, revenue-weighting causes the pooled dispersion to be larger in the unweighted calculations. Second, the unweighted calculations give relatively more weight, within indus-

	Dispersion of $tfpq$			Dispersion of ϕ			Percent Decline		
Sample	90/10	75/25	SD	90/10	75/25	SD	90/10	75/25	SD
Milk-Bulk	0.809	0.306	0.316	0.681	0.322	0.303	20.6%*	-4.8%	4.4%
Flour	0.404	0.205	0.163	0.396	0.176	0.172	2.0%	17.9%*	-5.3%
Concrete	0.558	0.275	0.238	0.522	0.260	0.230	7.1%*	5.8%*	3.4%*
Yarn	0.620	0.310	0.262	0.629	0.312	0.248	-1.4%	-0.6%	5.6%*
Boxes, Year$\leq$1987	0.475	0.204	0.199	0.409	0.196	0.185	17.3%*	4.1%*	8.1%*
Gasoline	0.309	0.151	0.147	0.296	0.137	0.141	4.6%	10.7%	4.8%
Coffee	0.635	0.321	0.257	0.569	0.272	0.249	12.3%*	19.6%*	3.3%
Sugar	0.692	0.330	0.313	0.807	0.352	0.352	-13.2%*	-6.3%	-10.6%*
Boxes, Year$\geq$1992	0.617	0.318	0.242	0.548	0.278	0.221	13.4%*	15.7%*	10.1%*
Milk-Packaged	0.564	0.284	0.235	0.531	0.262	0.226	6.5%*	8.9%*	4.2%*
Pooled-Bench.	0.527	0.253	0.227	0.493	0.238	0.219	7.2%*	6.5%*	3.9%*

Table 28: Dispersion of $tfpq$ and ϕ. Obvservations are not weighted. In the final three columns, stars indicate that he difference between $tfpq$ and ϕ is statistically significant, at the 5% level (see Appendix E for details). Due to Census' rules regarding data confidentiality, I am prohibited from reporting the actual quantiles of any empirical distribution. See the caption of Table 3 for a description of the imputation of the quantiles of $tfpq$ and ϕ.

tries, to the low-productivity, low employment plants, again causing unweighted dispersions to be larger than the weighted dispersions.

The decline in dispersion, of the pooled distribution of productivities, is smaller when observations are not weighted by revenue. For example, compared to the 8.8% decline that is given in Table 3, the 90/10 ratio of $tfpq$ is only 7.2% larger than the 90/10 ratio of ϕ. The difference, between the unweighted and weighted calculations, is due to differences in the weight that particular industries get. When observations are not weighted by revenue, the ready-mix concrete industry (which had a particularly small decline in productivity dispersion) is relatively more important in the calculations. On the other hand, when observations are weighted by revenue, the gasoline industry (which has a slightly larger than average decline in productivity dispersion) is relatively more important in the calculations. Note that, weighting by revenue, does not cause the within-industry declines in dispersion to be systematically larger or smaller. For the sample of industries with substantial variation in output quality, there are no systematic differences between the weighted and unweighted calculations (compare Tables 19 and 29).

Sample	Dispersion of $tfpq$			Dispersion of ϕ			Percent Decline			
	90/10	75/25	SD	90/10	75/25	SD	90/10	75/25	SD	N
Wine	1.407	0.735	0.502	1.444	0.779	0.536	-2.6%	-5.4%	-6.1%	330
Softwood Cut S.	1.379	0.675	0.490	1.310	0.651	0.489	5.4%	3.6%	0.2%	160
Pickles	0.890	0.441	0.346	0.962	0.471	0.368	-7.3%	-6.2%	-5.8%	145
Sausages	0.800	0.409	0.316	0.730	0.358	0.308	10.0%*	15.3%*	2.5%	621
Pooled-Quality	1.028	0.500	0.400	1.021	0.477	0.410	0.7%	4.8%	-2.5%	1256

Table 29: Dispersion of $tfpq$ and ϕ. Observations are not weighted. Stars indicate that he difference between $tfpq$ and ϕ is statistically significant, at the 5% level (see Appendix E for details). Due to Census' rules regarding data confidentiality, I am prohibited from reporting the actual quantiles of any empirical distribution. See the caption of Table 3 for a description of the imputation of the quantiles of $tfpq$ and ϕ.

D.7 Correcting for sample selection in productivity decompositions

As mentioned in Section 3.2, plants in the benchmark sample tend to exit and enter less frequently, compared to plants from their corresponding industries. As a result, the productivity decompositions of Section 4.5 may underrepresent the role of entry and exit in generating aggregate productivity growth. In this subsection, I try to account for this sample selection problem.

Table 30 presents the aggregate productivity growth decompositions, corrected for the underrepresentation of entering and exiting plants in the benchmark sample. For each industry in my benchmark sample, I compute the corrected Entry (Exit) Effects by dividing by the ratio of the sales-weighted-fraction of entrants (exiting plants) in the overall sample to the sales-weighted-fraction of entrants (exiting plants) in the benchmark sample. The correction that I make will magnify the share of entrants/exiting plants to the extent that entrants/exiting plants are underrepresented in the benchmark sample. Specifically, the corrected Entry and Exit Effects are given by (FHK denotes the decomposition method of Foster, Haltiwanger, and Krizan (2001), while GR denotes the decomposition method of Griliches and Regev (1995).):

$$\begin{aligned} \text{Entry Effect}_{FHK} &= \frac{\Pr\{i \in \mathcal{N} \mid i \in \text{overall sample}\}}{\Pr\{i \in \mathcal{N} \mid i \in \text{benchmark sample}\}} \\ &\times \sum_{i \in \mathcal{N} \cap \text{benchmark}} \theta_{i,t-1}(tfp_{it} - \overline{tfp}_{t-1}) \end{aligned} \tag{54}$$

Productivity Measure	Weight Industries By:	Total	Entry	Exit	Net Entry	Entry	Exit	Net Entry
$tfpr$	Real Revenues	-1.60	-0.06	0.11	0.04	-0.06	0.17	0.11
$tfpq$	Real Revenues	-1.60	-0.08	0.20	0.12	-0.07	0.26	0.19
ϕ	Real Revenues	-1.60	-0.12	0.17	0.05	-0.11	0.23	0.12
$tfpr$	# of Plants	1.30	0.30	0.17	0.47	0.20	0.28	0.49
$tfpq$	# of Plants	1.30	0.40	0.15	0.54	0.30	0.26	0.56
ϕ	# of Plants	1.30	0.27*	0.18	0.45	0.18*	0.29	0.47
Decomposition Method			Foster et. al			Griliches and Regev		

Table 30: Aggregate productivity growth decompositions, using the measures given in Equations 54- 57. All values are percentages. When $tfpr$ or ϕ is the productivity measure, stars indicate that the value given in the cell is significantly different than the corresponding value that uses $tfpq$ as the measure of plant productivity. See Appendix E for a detailed description of the bootstrapping procedure.

$$\begin{aligned} \text{Exit Effect}_{FHK} &= -\frac{\Pr\{i \in \mathcal{X} \mid i \in \text{overall sample}\}}{\Pr\{i \in \mathcal{X} \mid i \in \text{benchmark sample}\}} \\ &\quad \times \sum_{i \in \mathcal{X} \cap \text{benchmark}} \theta_{i,t-1}(tfp_{i,t-1} - \overline{tfp}_{t-1}) \end{aligned} \tag{55}$$

$$\begin{aligned} \text{Entry Effect}_{GR} &= \frac{\Pr\{i \in \mathcal{N} \mid i \in \text{overall sample}\}}{\Pr\{i \in \mathcal{N} \mid i \in \text{benchmark sample}\}} \\ &\quad \times \sum_{i \in \mathcal{N} \cap \text{benchmark}} \theta_{it}\left(tfp_{it} - \frac{1}{2}\overline{tfp}_{t-1} - \frac{1}{2}\overline{tfp}_{t}\right) \end{aligned} \tag{56}$$

$$\begin{aligned} \text{Exit Effect}_{GR} &= -\frac{\Pr\{i \in \mathcal{X} \mid i \in \text{overall sample}\}}{\Pr\{i \in \mathcal{X} \mid i \in \text{benchmark sample}\}} \\ &\quad \times \sum_{i \in \mathcal{X} \cap \text{benchmark}} \theta_{i,t-1}\left(tfp_{i,t-1} - \frac{1}{2}\overline{tfp}_{t-1} - \frac{1}{2}\overline{tfp}_{t}\right) \end{aligned} \tag{57}$$

As in Table 8, I average over the industries in the benchmark sample to arrive at an aggregate Entry Effect, Exit Effect, and Net Entry Effect. The Net Entry Effect is less than 0.1 percentage points larger after correcting for the underrepresentation of entering and exiting plants in the benchmark sample. As in Table 8, the only statistically significant difference among the three productivity measures is that the role of Entry is lager when $tfpq$, instead of ϕ, is used as the productivity measure.

Sample	Boxes	Concrete	Pooled	Boxes	Concrete	Pooled
$\overline{tfpq_{it}}$	-0.261*	-0.183*	-0.245*	-0.252*	-0.185	-0.236*
	(0.060)	(0.093)	(0.050)	(0.058)	(0.115)	(0.049)
$\bar{p}_{it}^{in,local}$	0.008	0.123	0.004	0.012	0.032	0.003
	(0.030)	(0.092)	(0.021)	(0.030)	(0.109)	(0.020)
N	190	131	321	190	131	321
Adjusted R^2	0.099	0.046	0.084	0.094	0.078	0.088
Division F.E.?	No	No	No	Yes	Yes	Yes

Table 31: Coefficient estimates and robust standard errors, from the regressions defined by Equation 34, with the addition of $\bar{p}_{it}^{in,local}$ as an explanatory variable. Stars indicate significance at the 5% level.

D.8 Including local prices in regressions 34, 35 and 40

One concern, regarding the regressions corresponding to Equations 34, 35 and 40, is that division fixed effects may not sufficiently control for the geographic forces that generate variation in $tfpq_{it}$. Unfortunately, since there are so few observations in the sample of corrugated box and concrete manufacturers, I cannot include fixed effects of greater geographic detail. Instead, to control for the geographic factors that generate variation in $tfpq_{it}$, I include–on the right-hand side of Equations 34, 35 and 40–the average materials price paid by plants that are close to plant i. In particular, I define $\bar{p}_{it}^{in,local}$ as the logarithm of the average (value-weighted) price paid by all of the establishments, other than i, that are located less than 50 miles from plant i. Materials prices are spatially correlated for concrete, but not for boxes (i.e., p_{it}^{in} is correlated to $\bar{p}_{it}^{in,local}$ only for the subsample of concrete manufacturers).

Regressions of plants' materials prices on suppliers' marginal costs are given in Table 31. The estimated coefficient corresponding to $\bar{p}_{it}^{in,local}$ is not significantly greater than 0, and tends to be somewhat larger for the subsample of ready-mix concrete manufacturers. Importantly, the coefficient estimates of the $\overline{tfpq_{it}}$ term are unchanged after including $\bar{p}_{it}^{in,local}$ as an explanatory variable.

In Table 32, I present the regression results corresponding to Equation 35, while in Table 33 I present the regression results corresponding to Equation 40. Again, the coefficient estimates of $\overline{tfpq_{it}}$ are ψ_{it} are not significantly altered by the inclusion of local prices, $\bar{p}_{it}^{in,local}$, as an explanatory variable. Also, the coefficient estimate of $\bar{p}_{it}^{in,local}$ is significant only for the subsample of ready-mix concrete manufacturers for one of the specifications.

Sample	Boxes	Concrete	Pooled	Boxes	Concrete	Pooled
$\overline{tfpq}_{it}$	0.078	0.133*	0.090	0.028	0.128*	0.070
	(0.066)	(0.046)	(0.052)	(0.062)	(0.055)	(0.047)
$\bar{p}_{it}^{in,local}$	0.029	0.021	0.018	0.009	-0.049	0.014
	(0.030)	(0.037)	(0.018)	(0.030)	(0.055)	(0.018)
N	190	131	321	190	131	321
Adjusted R^2	0.001	0.053	0.010	0.029	0.050	0.040
Division F.E.?	No	No	No	Yes	Yes	Yes

Table 32: Coefficient estimates and robust standard errors, from the regressions defined by Equation 35, with the addition of $\bar{p}_{it}^{in,local}$ as an explanatory variable. Stars indicate significance at the 5% level.

Sample	Boxes	Concrete	Pooled	Boxes	Concrete	Pooled
$\overline{tfpq}_{it}$	-0.231*	-0.129	-0.209*	-0.227*	-0.128	-0.207*
	(0.097)	(0.083)	(0.052)	(0.056)	(0.103)	(0.048)
$\bar{p}_{it}^{in,local}$	0.011	0.169*	0.006	0.018	0.139	0.007
	(0.028)	(0.070)	(0.020)	(0.028)	(0.090)	(0.019)
ψ_{it}	0.322*	0.886*	0.422*	0.328*	0.928*	0.420*
	(0.097)	(0.266)	(0.123)	(0.105)	(0.224)	(0.126)
N	190	131	321	190	131	321
Adjusted R^2	0.180	0.455	0.214	0.179	0.518	0.217
Division F.E.?	No	No	No	Yes	Yes	Yes

Table 33: Coefficient estimates and robust standard errors, from the regressions defined by Equation 40, with the addition of $\bar{p}_{it}^{in,local}$ as an explanatory variable. Stars indicate significance at the 5% level.

E Details of the Bootstrapping Exercises

The purpose of this section is to describe, and give the results of, the four bootstrapping exercises that are employed in Sections 4 and 5. The four bootstrapping exercises are used to determine a) whether the correlations among certain plant-level statistics are significantly different from 0, b) whether the dispersion of $tfpq$ is different from that of ϕ or $tfpr$, c) whether the Entry/Exit/Net Entry Effects (as in Equations 27-28) are significantly different when $tfpq$ is used instead of ϕ or $tfpr$, d) whether the declines in dispersion that are reported in Tables 10, 12, and 15 are significantly more than would be expected by simply adding independent variables. Below, I explain how each of the bootstrapping exercises is performed, and give the resulting confidence intervals.

To determine whether specific correlations among plant-level statistics are different from 0, I take 1000 bootstrapped sample, from the benchmark sample of 10,503 plant-year observations (or 1256 observations in the case of the Output Quality Variation sample). In each bootstrapped sample, the number of plants taken from each industry-year is the same as the benchmark sample. After sampling, I de-mean, as in Equation 23, and then compute the weighted and unweighted correlations. The 95% confidence intervals are provided in Tables 34, and 35.[60]

I follow a similar procedure to determine whether $tfpq$ is significantly more disperse than $tfpr$ or ϕ: For each (out of 1000) bootstrapped sample, I de-mean plant-level statistics, as in Equation 23, and then compute dispersions (the standard deviations, the 90/10 ratios, and the 75/25 ratios) of $tfpq$, $tfpr$, and ϕ. I then take the ratio of the dispersion of $tfpq$ and the dispersion of either $tfpr$ or ϕ. The 95% confidence intervals are provided in Table 36. In most cases, the left endpoint of the confidence interval is greater than 1, meaning that $tfpq$ is significantly more disperse than both $tpfr$ and ϕ. For the benchmark, pooled sample, $tfpq$ is significantly more disperse than ϕ, except when observations are revenue-weighted and the interquartile range is the measure of dispersion. The same can be said for the dispersions of $tfpq$ and $tfpr$.

And again, I follow a similar procedure to determine whether the Entry/Exit/Net Entry Effects (as in Equations 27-28) are significantly different when $tfpq$ is used instead of ϕ or $tfpr$. Again, I take 1000 bootstrapped samples, where, in each bootstrapped sample, the number of plants taken from each industry-year is the same as in the benchmark sample. For each boostrapped sample, I compute the Entry, Exit, and Net Entry Terms, by plugging in $tfpq$, $tfpr$, and ϕ into Equations 27 and 28. I then compute the difference, between

[60] Throughout this section, the confidence intervals correspond to the revenue-weighted calculations. Confidence intervals corresponding to the unweighted calculations are available upon request.

Sample	$\rho(p^{in},tfpq)$	$\rho(p^{in},\phi)$	$\rho(p^{in},p^{out})$	$\rho(\ \phi,tfpq)$	$\rho(\ tfpq,tfpr)$	$\rho(\ tfpq,p^{out})$
Milk-Bulk	-0.68,-0.13	-0.44,0.14	0.06,0.70	0.93,0.98	0.26,0.67	-0.89,-0.56
Flour	-0.48,-0.31	0.34,0.57	0.18,0.43	0.54,0.72	0.00,0.27	-0.78,-0.66
Concrete	-0.36,-0.26	0.07,0.17	0.23,0.33	0.89,0.92	0.71,0.77	-0.52,-0.43
Yarn	-0.45,-0.17	-0.05,0.40	0.08,0.36	0.82,0.92	0.36,0.61	-0.84,-0.67
Boxes, Yr.$\leq$'87	-0.36,-0.21	0.14,0.31	0.21,0.35	0.83,0.90	0.37,0.48	-0.83,-0.76
Gasoline	-0.48,-0.31	0.02,0.26	0.06,0.26	0.82,0.89	0.77,0.87	-0.48,-0.27
Coffee	-0.69,-0.20	-0.20,0.29	0.10,0.57	0.77,0.92	0.38,0.74	-0.73,-0.41
Sugar	-0.22,0.14	0.32,0.61	0.03,0.40	0.78,0.92	0.76,0.93	-0.58,-0.32
Boxes, Yr.$\geq$'92	-0.43,-0.27	0.00,0.18	0.20,0.37	0.88,0.92	0.03,0.20	-0.90,-0.85
Milk-Packaged	-0.33,-0.23	-0.02,0.11	0.17,0.28	0.93,0.95	0.37,0.49	-0.80,-0.69
Pooled-Bench.	-0.42,-0.31	0.05,0.20	0.18,0.28	0.85,0.89	0.65,0.74	-0.60,-0.50
Wine	-0.47,-0.12	0.08,0.50	0.11,0.52	0.71,0.86	0.00,0.48	-0.89,-0.69
Softwood Cut	-0.66,-0.23	-0.32,0.19	0.22,0.66	0.87,0.94	0.15,0.56	-0.97,-0.89
Pickles	-0.41,0.05	-0.14,0.35	-0.13,0.31	0.93,0.97	0.29,0.60	-0.84,-0.61
Sausages	-0.33,-0.04	0.12,0.39	0.06,0.35	0.87,0.92	0.10,0.44	-0.84,-0.73
Pooled-Quality	-0.38,-0.17	0.14,0.36	0.16,0.41	0.82,0.89	0.16,0.42	-0.86,-0.75

Table 34: Confidence intervals of correlations among plant-level characteristics. See text for details.

Benchmark Sample:

	p^{in}	p^{out}	$tfpq$	ϕ
p^{out}	0.180, 0.279			
$tfpq$	-0.424,-0.314	-0.599,-0.504		
ϕ	0.054,0.200	-0.518,-0.422	0.852,0.890	
$tfpr$	-0.303,-0.163	0.149,0.284	0.652,0.736	0.568,0.660

Output Quality Variation Sample:

	p^{in}	p^{out}	$tfpq$	ϕ
p^{out}	0.160,0.407			
$tfpq$	-0.385,-0.166	-0.858,-0.165		
ϕ	0.137,0.363	-0.733,-0.560	0.819,0.888	
$tfpr$	-0.143,0.180	0.241,0.423	0.157,0.418	0.189,0.404

Table 35: Confidence intervals of correlations among plant-level characteristics. See text for details.

Sample	Ratio of dispersion of $tfpq$ to dispersion of ϕ			Ratio of dispersion of $tfpq$ to dispersion of $tfpr$		
	90/10	75/25	SD	90/10	75/25	SD
Milk-Bulk	0.970,1.401	0.854,1.498	0.970,1.273	0.977,6.031	0.795,4.583	1.166,3.287
Flour	0.927,1.180	1.040,1.438	0.927,1.059	1.079,1.892	1.232,1.637	0.992,1.463
Concrete	1.037,1.118	1.015,1.126	1.037,1.069	1.047,1.176	1.026,1.165	1.072,1.154
Yarn	0.824,1.028	0.770,1.027	0.824,1.109	1.220,2.101	1.118,2.017	1.341,2.239
Boxes,Yr.$\leq$'87	1.002,1.078	0.972,1.058	1.002,1.060	1.573,1.757	1.427,1.908	1.654,2.098
Gasoline	0.976,1.170	0.974,1.227	0.976,1.145	0.931,1.146	0.911,1.137	0.980,1.125
Coffee	1.018,1.783	1.012,1.671	1.018,1.406	0.894,2.182	0.903,2.016	0.947,1.590
Sugar	0.710,1.018	0.689,1.233	0.710,0.962	0.931,1.364	0.834,1.443	1.030,1.275
Boxes,Yr.$\geq$'92	1.022,1.146	1.008,1.123	1.022,1.117	2.534,4.309	2.517,3.853	2.024,3.230
Packaged Milk	1.036,1.110	1.009,1.113	1.036,1.062	1.485,2.072	1.562,1.942	1.375,1.863
Pooled-Weighted	1.002,1.168	0.996,1.200	1.002,1.104	1.066,1.227	0.984,1.281	1.123,1.249
Pooled-Unweighted	1.056,1.091	1.050,1.093	1.028,1.050	1.415,1.515	1.474,1.600	1.303,1.374

Table 36: Confidence intervals of a) the ratio of the dispersion of $tfpq$ to the dispersion of $tfpr$–given in the left three columns–and b) the ratio of the dispersion of $tfpq$ to the dispersion of ϕ–given in the right three columns. See text for details.

Measure	Weight Ind. By:	Entry	Exit	Net Entry	Entry	Exit	Net Entry
$tfpr$	Sales	-0.06,0.11	-0.05,0.19	-0.07,0.23	-0.06,0.11	-0.06,0.19	-0.07,0.24
ϕ	Sales	-0.02,0.09	-0.06,0.10	-0.05,0.14	-0.02,0.09	-0.07,0.10	-0.05,0.13
$tfpr$	Plants	-0.05,0.23	-0.12,0.13	-0.09,0.28	-0.04,0.23	-0.12,0.13	-0.09,0.28
ϕ	Plants	0.00,0.16	-0.09,0.05	-0.05,0.15	0.01,0.15	-0.10,0.06	-0.04,0.15
Decomp. Method		Foster, Haltiwanger, and Krizan			Griliches and Regev		

Table 37: Confidence intervals of the difference, when $tfpq$, instead of $tfpr/\phi$, is used as the measure of plant productivity, of the Entry Effect, Exit Effect, and Net Entry Effect. These three effects are defined in Equations 27 and 28.

the Entry/Exit/Net Entry Effects when $tfpq$ is used and the Entry/Exit/Net Entry Effects when $tfpr$ (or ϕ) is used.

Table 37 gives the confidence intervals of the difference–between when $tfpq$ is used and when $tfpr/\phi$ is used–of the Entry/Exit/Net Entry Effects. In the first and the third rows, 0 lies within each and every confidence interval: the Entry/Exit/Net Entry Effects are not significantly different for revenue productivity versus quantity productivity. On the other hand, when industries are weighted by the number of plants, the Entry Effect is significantly greater when $tfpq$, instead of ϕ, is used as the measure of productivity.

I follow a somewhat different procedure to determine whether the estimated dispersion declines, as reported in Tables 10, 12, and 15, are significantly more than would be

Include Division Fixed Effects?		No	Yes	No	Yes
Include $\overline{tfpq_{it}}$?		No	No	Yes	Yes
Sample Size	Sample				
131	Concrete		0.3066,0.3676	0.3563,0.3737	0.3025,0.3660
190	Boxes		0.1844,0.1963	0.1937,0.1977	0.1834,0.1959
321	Pooled		0.2149,0.2228	0.2220,0.2240	0.2145,0.2224
131	Concrete		0.1884,0.2078	0.2035,0.2100	0.1874,0.2074
190	Boxes		0.2089,0.2225	0.2192,0.2245	0.2071,0.2222
321	Pooled		0.2073,0.2168	0.2151,0.2182	0.2067,0.2182
131	Concrete	0.3559,0.3737	0.3054,0.3666	0.3458,0.3736	0.3010,0.3653
190	Boxes	0.1937,0.1977	0.1830,0.1960	0.1924,0.1977	0.1817,0.1955
321	Pooled	0.2214,0.2240	0.2144,0.2224	0.2205,0.2240	0.2137,0.2222

Table 38: Confidence intervals. The first 3 rows give the confidence intervals corresponding to the dispersions given in Table 10. Rows 4-6 give the confidence intervals corresponding to the dispersions given in Table 12 . The final 3 rows give the confidence intervals corresponding to the dispersions given in Table 15.

expected by simply adding independent variables. I implement the following algorithm 1000 times:

From the sample of plant-year observations, I construct two new variables $\mathcal{P}\left(\tilde{p}_{it}^{in,CFS}\right)$ and $\mathcal{P}\left(tfpq_{it}\right)$. These two variables are constructed by randomly permuting $\tilde{p}_{it}^{in,CFS}$ (or $tfpq_{it}$) among the observations from a given industry-year. I then regress $\mathcal{P}\left(\tilde{p}_{it}^{in,CFS}\right)$ (or $\mathcal{P}\left(tfpq_{it}\right)$) against all the combinations of right-hand-side variables of the regressions given in Equations 34, 35, and 40. Following these regressions, I compute the revenue-weighted standard deviations of the residuals. These residuals are stored, for each iteration.

The 95% confidence intervals are presented in Table 38. Rows 1-3 give the confidence intervals corresponding to the regression of Equation 34; Rows 4-6 give the confidence intervals corresponding to the regression of Equation 35; and Rows 7-9 give the confidence intervals corresponding to the regression of Equation 40 (where the ψ_{it} variable is necessarily included on the right hand side). To make things concrete, consider the values given in the first row and penultimate column. To construct these two values, I repeatedly regress random permutations of $\mathcal{P}\left(\tilde{p}_{it}^{in,CFS}\right)$ against $\overline{tfpq_{it}}$, and then store the standard deviation of the residuals from each regression. The smaller value equals the 2.5-percentile standard deviation, and the larger value equals the 97.5-percentile standard deviation.

References

Acemoğlu, Daron, Vasco M. Carvalho, Asuman Ozdağlar, and Alireza Tahbaz-Salehi, 2011. "The Network Origins of Aggregate Fluctuations," mimeo.

Ackerberg, Daniel A., Kevin Caves, and Garth Frazer, 2006. "Structural Identification of Production Functions," mimeo.

Asplund, Marcus and Volker Nocke. "Firm Turnover in Imperfectly Competitive Markets." *Review of Economic Studies*, 73(2): 295-327.

Atalay, Enghin, Ali Hortaçsu, and Chad Syverson, 2012. "Why Do Firms Own Production Chains?" mimeo.

Baily, N. Martin, Charles Hulten, and David Campbell, 1992. "Dynamics in Manufacturing Plants," *Brookings Papers on Economic Activity: Microeconomics,* 1992, 187-249.

Balasubramanian, Natarajan and Jagadeesh Sivadasan, 2011. "What Happens When Firms Patent? New Evidence from U.S. Economic Census Data," *Review of Economics and Statistics*, 93(1): 126–146.

Bartelsman, Eric J., and Mark Doms, 2000. "Understanding Productivity: Lessons from Longitudinal Microdata," *Journal of Economic Literature*, 38(3): 569–594.

Bergin, Paul R., and Robert C. Feenstra. "Pass-Through of Exchange Rates and Competition between Floaters and Fixers," *Journal of Money, Credit, and Banking,* 41(1): 35-70.

Bernard, Andrew B. and J. Bradford Jensen, 1999. "Exceptional Exporter Performance: Cause, Effect, or Both?" *Journal of International Economics*, 47(1): 1–25.

Bernard, Andrew B. J. Bradford Jensen, and Peter Schott, 2006. "Transfer Pricing by U.S.-based Multinational Firms," mimeo.

Bloom, Nicholas, and John Van Reenen. 2010. "Why Do Management Practices Differ Across Firms and Countries?" *Journal of Economic Perspectives*, 24(1): 203–224.

Blundell, Richard, and Stephen Bond, 2000. "GMM Estimation with Persistent Panel Data: An Application to Production Functions," *Econometric Reviews*, 19(3): 321-340.

Chaney, Thomas, 2011. "The Gravity Equation in International Trade: An Explanation," mimeo.

Davis, Steven, Cheryl Grim and John Haltiwanger, 2008. "Productivity Dispersion and Input Prices: The Case of Electricity," CES Working Paper Series, #08-33.

de Fontenay, Catherine C., and Joshua S. Gans, 2007. "Bilateral Bargaining with Externalities," mimeo.

Dubé, Jean-Pierre, Günter J. Hitsch, and Peter E. Rossi, 2010. "State Dependence and

Alternative Explanations for Consumer Inertia," *RAND Journal of Economics,* 41(3): 417-445.

Eslava, Marcela, John Haltiwanger, Adriana Kugler, and Maurice Kugler, 2004. "The Effects of Structural Reforms on Productivity and Profitability Enhancing Reallocation: Evidence from Colombia," *Journal of Development Economics*, 75(2): 333-371.

Eslava, Marcela, John Haltiwanger, Adriana Kugler, and Maurice Kugler, 2009. "Trade Reforms and Market Selection: Evidence from Manufacturing Plants in Colombia," NBER Working Papers, 14935.

Foster, Lucia, John Haltiwanger, and C.J. Krizan, 2001. "Aggregate Productivity Growth: Lessons from Microeconomic Evidence," in New Developments in Productivity Analysis, University of Chicago Press.

Foster, Lucia, John Haltiwanger, and Chad Syverson, 2008. "Reallocation, Firm Turnover, and Efficiency: Selection on Productivity or Profitability?" *American Economic Review*: 98(1): 394–425.

Fox, Jeremy and Valerie Smeets, 2011. "Does Input Quality Drive Measured Differences in Firm Productivity?" *International Economic Review,* 52(4): 961-989.

Gorodnichenko, Yuriy, 2010. "Using Firm Optimization to Evaluate and Estimate Productivity and Returns to Scale," mimeo.

Griliches, Zvi and Haim Regev, 1995. "Firm Productivity in Israeli Industry: 1979-1988." *Journal of Econometrics*, 65(1): 175-203.

Halpern, László , Miklós Koren, and Adam Szeidl, 2011. "Imported Inputs and Productivity," mimeo.

Heckman, James. J., 1981. "Statistical Models for Discrete Panel Data," in C.F. Manski and D. McFadden, eds., Structural Analysis of Discrete Data with Econometric Applications.

Hsieh, Chang-Tai, and Peter J. Klenow. 2009. "Misallocation and Manufacturing TFP in China and India," *Quarterly Journal of Economics*, 124(4): 1403–1448.

Jones, Chad, 2011. "Intermediate Goods and Weak Links in the Theory of Economic Development," *American Economic Journal: Macroeconomics*, 3 (2): 1-28.

Katayama, Hajime, Shihua Lu, and James R. Tybout, 2009. "Firm-Level Productivity Studies: Illusions and a Solution," *International Journal of Industrial Organization*, 27(3): 403-413.

Kellogg, Ryan, 2011. "Learning by Drilling: Inter-Firm Learning and Relationship Persistence in the Texas Oilpatch," *Quarterly Journal of Economics,* 126(4):1961-2004.

Klump, Rainer, Peter McAdam, and Alpo William, 2011. "The Normalized CES Production Function: Theory and Empirics," ECB Working Paper #1294.

Kremer, Michael, 1993. "The O-Ring Theory of Economic Development," *Quarterly Journal*

of Economics, 108(3): 551-575.

Levinsohn, James and Amil Petrin, 2003. "Estimating Production Functions Using Inputs to Control for Unobservables," *Review of Economic Studies*, 70(2): 317-341.

Manea, Mihai, 2011. "Bargaining in Stationary Networks," *American Economic Review,* 101(5): 2042-2080.

Oberfield, Ezra, 2011. "Business Networks, Production Chains, and Productivity," mimeo.

Olley, G. Steven and Ariel Pakes, 1996. "The Dynamics of Productivity in the Telecommunications Equipment Industry," *Econometrica*, 64(6): 1263–1297.

Ornaghi, Carmine, 2006. "Assessing the Effects of Measurement Errors on the Estimation of Production Functions," *Journal of Applied Econometrics*, 21(6): 879–891.

Prescott, Edward C., 1998. "Needed: A Theory of Total Factor Productivity,"*International Economic Review*, 39 (3): 525–551.

Raval, Devesh, 2011. "Estimation of a CES Production Function with Factor Augmenting Technology," mimeo.

Roberts, Mark J. and Dylan Supina, 1996. "Output Price, Markups, and Producer Size," *European Economic Review*, 40(3-5): 909-921.

Sakellaris, Plutarchos and Daniel J. Wilson, 2004. "Quantifying Embodied Technical Change," *Review of Economic Dynamics,* 7(1): 1-26.

Syverson, Chad, 2004a. "Market Structure and Productivity: A Concrete Example," *Journal of Political Economy*, 112(6): 1181–1222.

Syverson, Chad, 2004b. "Product Substitutability and Productivity Dispersion," *Review of Economics and Statistics*, 86(2): 534–550.

Syverson, Chad, 2011. "What Determines Productivity?" *Journal of Economic Literature*, 49(2): 326–365.

U.S. Department of the Interior, 1989. *Cement Mineral Yearbook.*

U.S. Department of Transportation, Bureau of Transportation Statistics, 2009. Table 3-21: Average Freight Revenue Per Ton-mile.

van Biesebroeck, Johannes, 2008. "The Sensitivity of Productivity Estimates," *Journal of Business and Economic Statistics*, 26(3): 311-328.

White, T. Kirk, Jerome P. Reiter, and Amil Petrin, 2012. "Plant-level Productivity and Imputation of Missing Data in U.S. Census Manufacturing Data," NBER Working Paper #17816.